Linear failure and cubic convergence for surface evolution with relaxed minimal deformation

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Abstract

The finite element scheme for relaxed minimal-deformation surface evolution prescribes normal velocity while allowing tangential mesh motion. We establish contrasting results for its original exact-integral formulation with the discontinuous geometric unit normals of the current elements. For linear elements, one fixed smooth torus, a tangential rigid-rotation field, and an admissible alternating mesh family give a positive lower bound for the uniform-in-time L^2 error of the lifted flow map, on an interval where the discrete solution exists uniquely and remains nondegenerate. For cubic elements, every admissible mesh family converges for every smooth exact solution with smooth inverse on a fixed interval: the original discrete solution exists uniquely throughout that interval, and its lifted map error is at most $C_T h^3 \sqrt{1 + |\log h|}$. The cubic proof first controls shape error by a signed-distance-weighted relative volume and then estimates weakly tangent map error. These results partially solve the low-degree convergence question; the quadratic case remains unresolved.

1 Introduction

Tangential redistribution can improve a surface mesh without changing its normal velocity. For a parametrized method, however, the resulting flow map is itself part of the approximation. Its convergence requires more than control of the shape of the surface. The relaxed minimal-deformation formulation of Gao, Li and Tang [1] uses a harmonic-map relaxation from the initial surface to the evolving one. Its finite element scheme uses the geometric normal of the computed surface directly.

For smooth solutions whose flow maps and inverse maps remain smooth, Theorem 2.1 of the author manuscript of Gao, Li and Tang [1] proves a lifted, uniform-in-time L^2 error bound of order h^k for polynomial degree $k \geq 4$. Section 7 of that manuscript leaves convergence for $k = 1, 2, 3$ open. The question concerns the same position-gradient equation, exact fixed-reference integrals, and raw geometric normals, which can jump across current element edges. The theorem and section locators for [1] used here refer to the linked author manuscript.

We prove failure for degree one and convergence for degree three. For the negative result, the initial surface is a fixed smooth torus and the external field is one fixed small rigid rotation tangent to it. Thus the continuous relaxed minimal-deformation solution is stationary. An admissible shape-regular, quasi-uniform mesh family alternates two unequal azimuthal spacings. There are fixed $T_*, c_* > 0$ such that the original semidiscrete solutions exist uniquely with nondegenerate faces on $[0, T_*]$, while

$$\liminf_{h \to 0} \sup_{0 \leq t \leq T_*} \left\| X_h^\ell(t) - \mathrm{id} \right\|_{L^2(\Gamma^0)} \geq c_*.$$

The lift is the closest-point lift from the initial nodal surface. The construction is given in Theorem 2.2.

For the positive result, fix any smooth exact solution on $[0,T]$ with smooth inverse, and any admissible cubic nodal mesh family under the original uniform parametrization bounds. For all sufficiently small h, the unmodified semidiscrete solution exists uniquely with nondegenerate elements through T, and

$$\sup_{0\le t\le T}\left\|X_h^\ell(t)-X(t)\right\|_{L^2(\Gamma^0)}\le C_T h^3\sqrt{1+|\log h|}.$$

Theorem 6.1 retains every smooth-solution and mesh-family hypothesis of the original cubic question. The constant is independent of h. This paper gives a partial solution of AIM Problem 541: the quadratic case remains unresolved.

The linear counterexample separates map error from surface shape. The alternating species create a displacement proportional to the mesh width whose reference gradient can remain nonzero. A scalar label of the weak normal constraint records this displacement. At the initial torus its rate is $-\varepsilon\cos\theta/6$, where θ is the meridional angle and ε the fixed rotation amplitude. A common analytic interval and a time modulus for the mean velocity control the label through the fast initial layer. The original rotation energy then converts its fixed-time gradient effect into a positive L^2 map-error lower bound.

For cubic elements a single map-error estimate would mix this tangent drift with the error in surface shape. We instead use two estimates. A relative volume weighted by signed distance controls shape error. Its derivative is paired with a projection defect that retains the current area ratio, because the algorithm's weak constraint uses the fixed reference measure. The first area variation annihilates pure normal slopes, leaving a vanishing tangent-to-shape feedback. This gives a normal-error bound strong enough to close the tangent energy, with the logarithmic factor from a two-dimensional discrete Sobolev estimate. The resulting differential closeness permits continuation of the actual mixed system to the final time.

Section 2 gives the original equations and the linear mesh construction. Sections 3 and 4 control its weak operators and common existence interval; Section 5 proves linear failure. Section 6 treats general smooth data with cubic elements. All physical fields are real. Fourier arguments use their complexification, with conventions stated beside their use. All discrete integrals use the fixed initial surface exactly.

2 The torus and its triangulations

We use the position-gradient relaxed minimal-deformation equations

$$X_t=v,\qquad (v-U_{\mathrm{ext}}(X,t))\cdot(n\circ X)=0,\qquad v-\Delta_{\Gamma^0}X=\kappa(n\circ X),\qquad X(0)=\mathrm{id}. \tag{2.1}$$

The Laplace–Beltrami operator and all reference integrals are on the initial surface Γ^0. The unit normal n belongs to the current surface. Thus κ is a scalar constraint multiplier.

For degree k, let S_h be the continuous degree-k parametric space on a nodal interpolant Γ_h^0. The semidiscrete equations are exactly

$$X_{h,t}=v_h,\qquad X_h(0)=\mathrm{id}, \tag{2.2}$$

$$\int_{\Gamma_h^0}(v_h-U_{\mathrm{ext}}(X_h,t))\cdot N_h\,\chi_h=0\qquad(\chi_h\in S_h), \tag{2.3}$$

$$\int_{\Gamma_h^0}v_h\cdot\eta_h+\int_{\Gamma_h^0}\nabla_{\Gamma_h^0}X_h:\nabla_{\Gamma_h^0}\eta_h=\int_{\Gamma_h^0}\kappa_h N_h\cdot\eta_h\qquad(\eta_h\in S_h^3). \tag{2.4}$$

Here N_h is the pullback of the full geometric unit normal of each current element. It can jump across edges. All integrals are exact. In this section and Sections 3–5, we take $k=1$, so the reference elements are planar triangles. Section 6 uses the same equations for $k=3$.

2.1 Data and the mesh family

Fix $R = 2$, $r = 2/5$, let $\mathbb{T} = \mathbb{R}/(2\pi\mathbb{Z})$, and set

$$\Psi(\theta, \phi) = ((R + r\cos\theta)\cos\phi, (R + r\cos\theta)\sin\phi, r\sin\theta), \qquad (\theta, \phi) \in \mathbb{T}^2. \tag{2.5}$$

The orientation is outward. Write $\rho_\theta = R + r\cos\theta$, $\Omega x = (-x_2, x_1, 0)$, and take one fixed, sufficiently small $\varepsilon > 0$ with

$$U_{\mathrm{ext}}(x, t) = \varepsilon\Omega x. \tag{2.6}$$

This field is tangent to the torus. Its metric is $\mathrm{diag}(r^2, \rho_\theta^2)$, and direct differentiation gives

$$\Delta_{\Gamma^0}\Psi = -\left(\frac{1}{r} + \frac{\cos\theta}{\rho_\theta}\right) n.$$

Thus $X = \mathrm{id}$, $v = 0$, and $\kappa = 1/r + \cos\theta/\rho_\theta$ solve (2.1) smoothly on every fixed interval.

For $n \in \mathbb{N}$, put

$$\delta = \frac{2\pi}{4n}, \qquad s = 4\delta, \qquad \theta_i = is, \qquad a = \frac{1}{3}, \quad b = \frac{2}{3}.$$

The two azimuthal labels in cell j, $0 \le j < 4n$, are

$$\phi_{j,+} = j\delta - a\delta/2, \qquad \phi_{j,-} = j\delta + a\delta/2.$$

Take nodal values of Ψ at these angles. In two neighboring rows denote the nodes in one cell by $A = (i, +)$, $B = (i, -)$, $C = (i + 1, +)$, $D = (i + 1, -)$. Primes indicate the next azimuthal cell. Use the oriented faces

$$ABD, \qquad ADC, \qquad BA'C', \qquad BC'D. \tag{2.7}$$

The short and long azimuthal widths are $a\delta$ and $b\delta$. Let Y_h be the identity on the resulting planar surface.

Lemma 2.1. *For sufficiently large n, these are admissible shape-regular, quasi-uniform degree-one nodal meshes, with $h \asymp \delta$. The closest-point projection $a_h : \Gamma_h^0 \to \Gamma^0$ is bijective, its face Jacobians are bounded above and below uniformly, and*

$$\left\| Y_h^\ell - \mathrm{id} \right\|_{L^\infty(\Gamma^0)} \le C\delta^2, \qquad Y_h^\ell = Y_h \circ a_h^{-1}.$$

Here K_f is the flat triangle with the same vertices as K. The degree-one element parametrization $F_K : K_f \to K$ is the identity.

Proof. The limiting edge directions are the nonparallel meridional and azimuthal tangents, of lengths rs, $\rho_\theta a\delta$, and $\rho_\theta b\delta$. Their normalized lengths and angles stay in compact nondegenerate sets. The distance and normal errors are $O(\delta^2)$ and $O(\delta)$, by trigonometric expansion.

The closest projection is explicit. Put $\rho_{xy} = (x_1^2 + x_2^2)^{1/2}$ and $d = ((\rho_{xy} - R)^2 + x_3^2)^{1/2}$. Then

$$a(x) = \left(\left[R + \frac{r(\rho_{xy} - R)}{d}\right]\frac{x_1}{\rho_{xy}}, \left[R + \frac{r(\rho_{xy} - R)}{d}\right]\frac{x_2}{\rho_{xy}}, \frac{rx_3}{d}\right).$$

Both denominators are uniformly positive near the meshes. Each azimuthal quadrilateral is planar. At fixed azimuth ϕ, its cylindrical radius is the affine interpolation of its two row radii multiplied by

$$c_\phi = \frac{\cos(d_\phi/2)}{\cos(\phi - \phi_{\mathrm{mid}})} = 1 + O(\delta^2),$$

where d_ϕ is its angular width. The determinant of the cross-section vector about $(R, 0)$ and its meridional chord is $r^2\sin s + O(\delta^2 s) > 0$. Its projected cross-section angle is strictly increasing, as is the azimuth. The lifted angular coordinates cover both periods once. This proves bijectivity. The tangent estimates give the Jacobian bounds and the identity error. □

Theorem 2.2. *For the fixed torus (2.5), one fixed field (2.6), and the mesh family (2.7), there are $T_* > 0$, $c_* > 0$, and n_0 such that the original system (2.2)–(2.4) has a unique solution with nondegenerate evolving elements on $[0, T_*]$ for $n \geq n_0$, but*

$$\liminf_{n\to\infty} \sup_{0\leq t\leq T_*} \left\| X_h^\ell(t) - \mathrm{id} \right\|_{L^2(\Gamma^0)} \geq c_*. \tag{2.8}$$

Here $X_h^\ell = X_h \circ a_h^{-1}$ is the closest-point lift and $h \asymp \delta = 2\pi/(4n)$. All constants and the field are fixed before the mesh limit.

The theorem concerns the parametrized flow map. Its proof needs a common existence interval: an initial velocity calculation alone would not exclude disappearance of the effect in an initial layer. We establish that interval in Section 4 and complete the proof in Section 5.

2.2 The full local equation and cyclic coordinates

Let M, K, and $T(X)$ denote the exact vector mass, reference stiffness, and weak normal matrices. In physical nodal coordinates,

$$X_t = V, \qquad MV + KX = T(X)^*\kappa, \qquad T(X)(V - \varepsilon\Omega X) = 0. \tag{2.9}$$

Initially $X = Y$, the nodal identity. These matrices depend smoothly on X while all scaled edge cross products stay nonzero.

Let N_{ref} insert the smooth reference nodal normal. On a small geometric neighborhood, every current face normal has a common positive component in this direction. Hence TN_{ref} is a definite weighted scalar mass plus $O(\delta)$: replacing a nodal reference normal by a common reference normal on each face costs $O(\delta)$ in its mass norm, and the remaining weights are positive. Its Hermitian part has a uniform lower bound. This argument holds on the full mesh, not only on cyclic modes. Thus T has full row rank and $TM^{-1}T^*$ is invertible. Eliminating κ gives a smooth finite-dimensional ODE. Picard iteration on a closed coordinate ball gives a local solution: a short time makes the integral map a contraction, and the same Lipschitz bound gives uniqueness.

Cyclic rotations preserve (2.9). Full uniqueness therefore gives the exact representation

$$X_{i,j,\pm} = \mathcal{R}_{\phi_{j,\pm}}(y_i \pm \delta w_i), \qquad y_i(0) = y_i^0 = (R + r\cos\theta_i, 0, r\sin\theta_i), \quad w_i(0) = 0, \tag{2.10}$$

where $\mathcal{R}_\phi$ is rotation about the third axis. All three Cartesian components of both profiles remain present.

3 The geometric constraint and its weak operators

3.1 Row norms and exact element formulas

The row inner product is $\langle p, q\rangle_0 = s\sum_i p_i^* q_i$. Let τ be the forward row permutation and set

$$D = \frac{\tau - I}{s}, \qquad D^* = \frac{\tau^{-1} - I}{s} = -\tau^{-1}D, \qquad \lambda_k = \frac{e^{iks} - 1}{s}.$$

Stars refer to this uniform row inner product. A varying-mass adjoint is written explicitly as $M^{-1}A^*M$. With $\widehat{q}_k = n^{-1}\sum_i q_i e^{-ik\theta_i}$, define

$$\|q\|_{H^\sigma}^2 = 2\pi \sum_k (1 + |\lambda_k|^2)^\sigma |\widehat{q}_k|^2. \tag{3.1}$$

Signed modular frequencies are used. For $|k| \le n/2$, $|\lambda_k| \ge 2|k|/\pi$, and $|\lambda_{k+l}| \le |\lambda_k| + |\lambda_l|$, including aliasing. Fourier convolution and Cauchy–Schwarz give the uniform H^1 embedding into ℓ^∞ and one-dimensional product estimates.

For a physical species vector write $q_\pm = q_b \pm q_o$, with label rotations understood, and put $q_c = q_o/\delta$. Its reference squared piecewise affine H^1 norm is uniformly equivalent to

$$\|q_b\|_{H^1}^2 + \|q_c\|_0^2 + \delta^2 \|Dq_c\|_0^2 \asymp \|q_b\|_{H^1}^2 + \|q_c\|_0^2 .$$

Here $\delta \|D\| \le 1/2$. We shall use the bases

$$\mathcal{X}_0 = H^4 \times H^3, \quad \mathcal{X}_1 = H^5 \times H^4, \qquad \mathcal{A}_0 = H^3 \times H^2, \quad \mathcal{A}_1 = H^4 \times H^3. \tag{3.2}$$

The macro component is the mean of the two nodal species; the optical component is half their difference. The scaled optical coordinate divides that difference by δ. Geometry is $\mathbf{e} = (y - y^0, w)$, so its physical error has mean $y - y^0$ and optical part δw. The velocity spaces have one fewer row derivative than the corresponding geometry spaces; each scaled optical component has one fewer than its mean.

Fix $p = 4$. For a row space B and $\rho > 0$, write

$$\varpi_j = \frac{(j+1)^p \rho^j}{j!}, \qquad \|q\|_{\rho,p;B}^2 = \sum_{j \ge 0} \varpi_j^2 \left\| D^j q \right\|_B^2 . \tag{3.3}$$

Coefficient norms have the same weights with coefficient row differences.

There are $4n$ azimuthal copies. Multiply each assembled one-cell weak matrix by $4n/s = (\pi/2)\delta^{-2}$. In the row inner product this represents exactly the full physical cyclic energy and leaves the equations unchanged. Thus normalized M, T are order zero. On a triangle of reference area A, the scalar mass is $A/12$ times the matrix with diagonal 2 and off-diagonal 1. The stiffness is $K = J^* J$, where J is the exact reference gradient including area and metric factors.

For scaled current edges $g_l = (X_l - X_0)/\delta$, put $c = g_1 \times g_2$, $n = c/|c|$, and $d_l = (h_l - h_0)/\delta$. The exact normal derivative is

$$DN[h] = \frac{(I - nn^{\mathsf{T}})(d_1 \times g_2 + g_1 \times d_2)}{|c|}. \tag{3.4}$$

We keep this full normal map. Its coefficients are analytic on a compact nondegenerate set of scaled edges. Rotation remainders retain their powers of δ by

$$\frac{\mathcal{R}_{\beta\delta} - I}{\delta} = \beta \int_0^1 \Omega \mathcal{R}_{t\beta\delta} \, dt.$$

3.2 Row moments and the normal graph

For $(Aq)_i = \sum_k a_i(k) q_{i+k}$, put $m_\ell(A)_i = \sum_k k^\ell a_i(k)$. All kernels below have exponentially summable distance moments and bounded coefficient differences at the displayed regularities. Class 0 imposes no zero condition; classes 1 and 2 mean

$$m_0 = \delta c_0, \qquad \text{and} \qquad m_0 = \delta^2 c_0, \quad m_1 = \delta c_1,$$

respectively. The factors remain present after coefficient differences. Physical first and second orders are δ^{-1} times class 1 and δ^{-2} times class 2.

Exactly, without commuting matrices,

$$\begin{aligned} m_0(AB)_i &= \sum_k a_i(k) m_0(B)_{i+k}, \\ m_1(AB)_i &= \sum_k k a_i(k) m_0(B)_{i+k} + \sum_k a_i(k) m_1(B)_{i+k}. \end{aligned} \tag{3.5}$$

Subtract the value of each right coefficient at i. A difference supplies $s = 4\delta$. Thus two class 1 factors give class 2. For a class 2 factor next to class 0, expand once more: the first moment contributes δs, and the second-difference remainder contributes s^2. Class 2 is preserved in either order.

The adjoint moments are

$$m_0(A^*)_i = \sum_l a_{i-l}(l)^*, \qquad m_1(A^*)_i = -\sum_l l\, a_{i-l}(l)^*.$$

The finite Laurent identity $\tau^{-l} = I - l(\tau - I) + (\tau - I)^2 R_l(\tau)$, with $\|R_l\|_{\ell^1} \le C(1+|l|)^2$, yields

$$m_0(A^*) = m_0(A)^* - sD_{\mathrm{coeff}}(m_1(A)^*) + s^2 \sum_l R_l(\tau) D^2_{\mathrm{coeff}}(a(l)^*). \tag{3.6}$$

Its minus sign corresponds to our forward difference. The analogous first-moment identity proves adjoint stability of the classes. Periodic moments are read on lifted periodized kernels; wrapping tails are exponential and smaller than each fixed required power of δ.

A uniformly invertible localized order-zero matrix has a localized inverse. Conjugate by $W_j = \exp(\gamma\ \mathrm{dist}(i,j))$. Weighted kernel moments give $\left\|WAW^{-1} - A\right\| \le C\gamma$, so a fixed small γ preserves invertibility by a Neumann series. One-row inputs then give exponential kernel decay for A^{-1}. If $A^+ = \tau A \tau^{-1}$ and $\delta_c A = (A^+ - A)/s$, exactly

$$\delta_c(A^{-1}) = -(A^+)^{-1}(\delta_c A)A^{-1}. \tag{3.7}$$

This also supplies the coefficient differences of the global inverse.

The actual mass-normal projection is

$$P_N = M^{-1}T^*(TM^{-1}T^*)^{-1}T, \qquad Q = I - P_N. \tag{3.8}$$

The full normal-column gap supplies its inverses. Freezing at output row i changes each kernel by $C\delta(1 + \mathrm{dist})$ times an exponential. At zero row phase the coherent frozen off-diagonal blocks vanish, since the alternating affine function on a whole rectangle satisfies $\int \chi = 0$, $\int \chi^2 = A_{\mathrm{rect}}/3$. A zero-sum cross kernel acts on neighboring differences and supplies the factor δ required by the scaled optical norm. More explicitly, in a coherent frozen strip let ζ be the row Fourier phase and set $d_\delta(\zeta) = (1 + |\zeta - 1|^2/s^2)^{1/2}$. The full physical state consists of both three-component species values, with mean and optical parts as above. The alternating affine function χ has nodal values $+1$ and -1 on the two species. At $\zeta = 1$, its rectangle integrals make both the mass and weak-normal matrices block diagonal in mean/optical coordinates. The Schur inverse therefore has the same zero cross blocks, and smooth dependence gives $|P_{bo}(\zeta)| + |P_{ob}(\zeta)| \le C|\zeta - 1|$. Conjugating by $\mathrm{diag}(d_\delta I_3, \delta^{-1} I_3)$ is bounded, since

$$\frac{|\zeta - 1|}{\delta d_\delta} \le C, \qquad \delta d_\delta |\zeta - 1| \le C.$$

The $O(\delta)$ actual-kernel freezing remainder is bounded before dividing the optical coordinate by δ. This gives a bounded order-zero remainder in the same mixed norm. The identity $[D, A^{-1}] = -A^{-1}[D, A]A^{-1}$ supplies differentiated bounds. In particular P_N, Q are uniformly bounded in macro H^1 plus scaled-optical L^2, and in their differentiated versions when the corresponding coefficient norms are retained.

Use the fixed aligned reference frame

$$e_\theta = (-\sin\theta, 0, \cos\theta), \quad e_\phi = (0,1,0), \quad n_0 = (\cos\theta, 0, \sin\theta).$$

Let C_{all} insert all four tangent coordinates $\mathbf{a} = (u, \omega)$, with physical optical size $\delta\omega$. Let L_{ref} extract them and let N_{ref} insert the reference normal coordinates. Set

$$R_N = N_{\mathrm{ref}}(TN_{\mathrm{ref}})^{-1}, \qquad \mathsf{E} = C_{\mathrm{all}} - R_N T C_{\mathrm{all}}. \tag{3.9}$$

Then $T\mathsf{E} = 0$, $L_{\mathrm{ref}}\mathsf{E} = I$, and $\mathsf{E}'[h] = -R_N T'[h]\mathsf{E}$. Every weak-tangent vector equals $\mathsf{E}L_{\mathrm{ref}}$ of itself. The global graph and its inverse have uniform mixed bounds. In scaled state coordinates their type is

$$\begin{pmatrix} \text{order } 0 & \delta^2 \text{ order } 1 \\ \text{order } 1 & \text{order } 0 \end{pmatrix}. \tag{3.10}$$

Consequently

$$\|\mathsf{E}\mathbf{a}\|_{\mathcal{X}_0} \le C_{\mathrm{low}}\big(\|\mathbf{a}\|_{\mathcal{A}_1} + \|\mathbf{e}\|_{\mathcal{X}_1}\|\mathbf{a}\|_{\mathcal{A}_0}\big). \tag{3.11}$$

Here $\mathcal{X}_0, \mathcal{X}_1$ are the geometry bases (H^4, H^3), (H^5, H^4), whereas $\mathcal{A}_0, \mathcal{A}_1$ are the velocity bases (H^3, H^2), (H^4, H^3). Low constants depend on fixed reference bounds and the stated current low norms, never on n. The high geometry factor is written explicitly.

3.3 The curved cancellation and the initial velocity

For a scalar multiplier define $H_\kappa h = T'[h]^*\kappa$. The alternating curvature scale is $\kappa_o = \lambda/\delta$. After removing the common factor δ^{-2}, the physical mean/optical blocks of $H_{\lambda/\delta}$ satisfy

	m_0	m_1	kernel size
bb	$O(\delta^2)$	$O(\delta)$	$O(1)$
bo, *ob*	$O(\delta)$	$O(1)$	$O(1)$
oo	$O(1)$	$O(1)$	$O(1)$.

(3.12)

Here the alternating signs are included in the multiplier, and all entries are factored identities with bounded smooth coefficients.

To prove them, freeze the lower-row divided meridional chord $p = (y_{i+1} - y_i)/\delta$, $q = \Omega y_i$, and w_i. The leading short and long rectangles have chords $p, aq - 2w_i$ and $p, bq + 2w_i$. They are planar, even when the two directions are not orthogonal. The alternating P1 function χ is affine azimuthally. A coarse row test L_ν is affine meridionally, and exact integration on the two triangles gives

$$\int_{\mathrm{rectangle}} \chi L_\nu = 0. \tag{3.13}$$

A constant aligned macro input changes the scaled edges first at order δ, through label rotations. That first variation is affine azimuthally, so its normal variation is constant on each leading rectangle. Equation (3.13) removes its order-δ pairing. Reference weights, neighboring multipliers, test rotations and normal coefficients then change it by another δ. This proves the *bb* zero sum. For the *bb* first moment the input is affine in row level; its leading normal variation is again constant on the whole rectangle. One curvature or coefficient difference remains. For *bo*, a constant optical input is affine azimuthally and the same integral vanishes. For *ob*, the constant leading macro input is a translation, so its normal variation is zero. The *oo* bound follows from (3.4).

These arguments factor the remainders by exact divided differences and the rotation integral. For example, the *bb* zero-sum coefficient uses macro first and second differences, optical first differences and multiplier first differences. A coefficient row difference raises each corresponding count by one and retains both mesh factors. Thus the conclusions apply to the actual curved triangles and their coefficient derivatives.

We need a distinct cancellation for the moving constraint. For $W_\pm = W_b \pm \delta W_c$, put $B_W h = M_{\mathrm{scalar}}^{-1} T'[h]W$. After removing δ^{-1}, its physical scalar-output/vector-input blocks have

	m_0	m_1
bb	$O(\delta)$	$O(1)$
bo	$O(1)$	$O(1)$
ob	$O(\delta^2)$	$O(\delta)$
oo	$O(\delta)$	$O(1)$.

(3.14)

The optical test at just one meridional row is important. On a short rectangle, a constant vector has optical test integral $A_{\mathrm{tri}}/3$ at the lower row and $-A_{\mathrm{tri}}/3$ at the upper row; on a long rectangle the signs reverse. Assembly is therefore a backward patch difference. A constant macro input already supplies its rotational factor δ, and this patch difference supplies the second factor in *ob*. An affine row input gives its first-moment zero. The other entries follow from the constant macro variation or the constant optical patch cancellation. The actual row-supported test has been used, rather than a bilinear product χL_ν. Scalar mass inversion preserves the table: its cross blocks have one zero, so a cross inverse followed by the *bb* single zero still has the *ob* double zero, by (3.5).

Laurent division by $\tau - I = sD$ now gives

$$(B_W h)_b : \ Dh_b, \ h_c, \qquad (B_W h)_o/\delta : \ D^2 h_b, \ Dh_c,$$

up to the stated lower coefficients. This is one derivative on the mixed scale. A mean multiplier gives the same first-order normal derivative estimate. An alternating *velocity* multiplier δl_c has two additional mesh factors relative to λ/δ, and is mixed order zero.

At $X = Y$, let $\kappa_s = 1/r + \cos\theta/\rho_\theta$. The reference force defect is $R_h = KY - T_Y^* \kappa_s$. In an orthonormal Monge chart the actual tangent-projected planar star annihilates affine nodal data exactly. Its curved metric differs from its projected planar metric by $O(\delta^2)$. The projected star differs from the leading product star by a relative $O(\delta)$, which acts only on quadratic data. For short/long lengths a_0, b_0 and meridional length c_0, the latter star has mass row sum $c_0(a_0 + b_0)/2$; its action on $(h_{11}x^2 + 2h_{12}xz + h_{22}z^2)/2$ is $-c_0(a_0 + b_0)(h_{11} + h_{22})/2$. Indeed its horizontal stiffness weights are $c_0/a_0, c_0/b_0$, its vertical weights are $(a_0 + b_0)/(2c_0)$, and its diagonal weights vanish. The affine part cancels on the actual projected star, including its perturbation. The metric error, perturbed quadratic weights and cubic remainder all cost $O(\delta^3)$ in weak force. Dividing by its $O(\delta^2)$ mass gives

$$\left\| M^{-1} R_h \right\|_{H^j} \le C_j \delta, \qquad \left\| M^{-1} KY \right\|_0 \le C. \tag{3.15}$$

Each fixed coefficient difference retains the order, since the reference descriptors are smooth periodic trigonometric functions. Their convergent angle series give the corresponding uniform positive analytic radius.

Since ΩY is tangent to the smooth torus, its weak normal projection is $O(\delta)$. Thus the actual initial velocity is

$$V(0) = -QM^{-1} R_h + P_N \varepsilon \Omega Y.$$

It is physically $O(\delta)$, its mean is $O(\delta)$ in H^1, and its optical coordinate divided by δ is uniformly bounded. We retain this bounded initial layer.

3.4 Force scales and the weak tangent form

Fix an analysis constant $\beta > 0$, let $e = X - Y$ in physical coordinates, and define

$$\widehat{\kappa} = (TN_{\mathrm{ref}})^{-*} N_{\mathrm{ref}}^* (KX + \beta M e), \qquad \widehat{B} = K + \beta M - H_{\widehat{\kappa}}. \tag{3.16}$$

Write $\widehat{\kappa} = (k_b, \lambda/\delta)$ in physical mean/optical multiplier coordinates. A physical primal pair is $(q_b, \delta q_c)$; pairing it with a force (F_b, F_o) gives $F_b \cdot q_b + \delta F_o \cdot q_c$. Thus the scaled dual force is $(F_b, \delta F_o)$, not $(F_b, F_o/\delta)$. The inverse normal map has the state type (3.10); its force-dual type is instead

$$\begin{pmatrix} \text{order } 0 & \text{order } 1 \\ \delta^2 \text{ order } 1 & \text{order } 0 \end{pmatrix}.$$

For $F = KX + \beta M e$, the row moment calculus gives

$$\|F_b\|_{H^2} + \|\delta F_o\|_{H^3} \le C_{\mathrm{low}}, \qquad \|k_b\|_{H^2} + \|\lambda\|_{H^3} \le C_{\mathrm{low}}.$$

The mean force uses D^2y, Dw, whereas δF_o uses Dy, w. The apparent extra order in the lower dual cross block is controlled by $\delta^2 D^2$. At the reference, (3.15) gives $\lambda = O(\delta^2)$. On an analytic geometry ball of radius η in $\mathcal{X}_0$,

$$\|\lambda\|_{H^3} \le C(\eta + \delta^2). \tag{3.17}$$

All these bounds hold in the row analytic norm (3.3). The scaled normal formulas and ordered inverse series are analytic there. Reference real-angle gaps are uniform by the same geometric argument for every common angle shift; a small complex shift preserves them. The holomorphic continuation of a real transpose is used, not conjugation of the complex shift. This gives a uniform reference series disk for the inverse coefficients.

Set $B = \mathsf{E}^* \widehat{B} \mathsf{E}$, with macro/fast blocks $B_{ss}, B_{sf}, B_{fs}, B_{ff}$. Their normalized classes are

$$\delta^{-2} \text{ class } 2, \quad \delta^{-1} \text{ class } 1, \quad \delta^{-1} \text{ class } 1, \quad \text{class } 0,$$

and hence

$$B_{ss} : H^3 \to H^1, \quad B_{sf} : H^2 \to H^1, \quad B_{fs} : H^3 \to H^2, \quad B_{ff} : H^2 \to H^2.$$

Products with E, its adjoint and global inverse kernels preserve the factored classes by (3.5) and (3.6). The alternating form is bounded by $C\,\|\lambda\|_{H^3}$ times the mixed H^1 norm squared. For the mean part (3.4) gives

$$|\,\langle v, H_{k_b} v\rangle_0\,| \le C\,\|k_b\|_\infty\,\|v\|_{L^2}\,\|v\|_{H^1}\,.$$

Choose β large from the reference mean bound, then η small relative to β, and finally δ small. The full tangent form B is uniformly coercive in the mixed H^1 norm. In particular $H := B_{ff}$ has a uniform gap and a localized global inverse.

The weak representation gives more information than these strong mapping bounds. Factoring the exact reference gradient before expanding a left adjoint gives

$$J\mathsf{E}_s = J_s D + J_0, \qquad J\mathsf{E}_f = J_f. \tag{3.18}$$

To obtain J_s, divide the optical-from-mean class 1 kernel by its phase difference. Its factored zero moment belongs to J_0. The macro part uses $D\mathsf{E}_s = \mathsf{E}_s D + [D, \mathsf{E}_s]$. For J_f, both the azimuthal gradient and δD are order zero. Thus J_s, J_f have coefficient H^3 from $\mathcal{X}_0$, and J_0 has H^2, using D^2y, Dw.

The nine stiffness coefficients are exactly

$$\begin{gathered} J_s^* J_s,\ J_s^* J_f,\ J_f^* J_s,\ J_f^* J_f, \\ J_s^* J_0,\ J_0^* J_s,\ J_0^* J_f,\ J_f^* J_0,\ J_0^* J_0. \end{gathered}$$

For the alternating multiplier, (3.12) places one D on the input and one D^* on the macro test. Moving the latter difference uses one coefficient difference. Its lower coefficients involve $D\lambda, D^2y, Dw, k_b$, and have H^2. The mean multiplier is a first-normal-derivative form; its principal gradient/fast terms retain an explicit small δ. The β mass is lower order, with any δD factor kept inside an order-zero kernel. Consequently the exact weak blocks can be written

$$\begin{aligned} B_{ss} &= D^* A D + D^* b_R + b_L D + c, \\ B_{sf} &= D^* C_f + d_f, \qquad B_{fs} = C_s D + d_s, \qquad B_{ff} = H. \end{aligned} \tag{3.19}$$

The principal coefficients A, C_f, C_s, H have H^3; the lower coefficients have H^2 at geometry $\mathcal{X}_0$. One additional lower-coefficient derivative costs exactly one $\mathcal{X}_1$ factor, linearly.

Lemma 3.1. *The actual global order-zero matrix* $\mathbb{A} = \begin{pmatrix} A & C_f \\ C_s & H \end{pmatrix}$ *has a uniformly positive Hermitian part on the small geometry ball.*

Proof. The coherent frozen reference form is the P1 Dirichlet form on an arbitrary macro gradient and fast coordinate. At zero phase the divided kernel is defined by its continuous extension after removing the macro phase zero; the class-1 factorization in (3.18) gives this extension. Its norm controls both: on the product strip, vanishing gradients force the optical coordinate and then the macro gradient to vanish. The same holds at zero phase after division by the macro phase difference. Compactness of phase and reference shapes gives a uniform gap. The small optical geometry and (3.17) preserve it; the mean-multiplier principal terms retain their small δ.

To pass from frozen to actual kernels, take a smooth row partition $\sum_\nu \chi_\nu^2 = 1$, with bounded overlap and windows of L rows. Kernel first moments give a commutator error C/L. Replacing coefficients in a window by its center's coherent frozen ones costs $CL\delta$; the inverse-kernel freezing error is included. Therefore

$$\operatorname{Re}\langle v, \mathbb{A}v\rangle_0 \geq [c - C(L\delta + 1/L)]\,\|v\|_0^2\,.$$

Choose the integer L once, then δ small. This proves the global gap. The localization length is a proof choice, not a discretization parameter. □

4 A uniform analytic interval

The analytic weights and norms are those in (3.3). The exact shifted Leibniz rule gives binomial convolution; the polynomial factor makes this an algebra. Indeed $(k+l+1)^p \leq C_p((k+1)^p + (l+1)^p)$, and $\sum(j+1)^{-2p} < \infty$. This proves the product bounds with uniform constants. Ordered inverse series and the binomial series for the full normal denominator converge in a sufficiently small geometry ball. Reference angle analyticity, uniform real gaps and complex perturbation give a positive initial radius independent of the mesh.

Proposition 4.1. *For the fixed sufficiently small* $\varepsilon > 0$*, there are* $T_0 > 0$*,* $\rho_* > 0$ *and* C*, independent of sufficiently small* δ*, such that the original solution exists uniquely with nondegenerate elements on* $[0, T_0]$*, and*

$$\sup_{t \leq T_0} \left(\|\mathbf{e}(t)\|_{\rho_*, p; \mathcal{X}_0} + \|(u(t), z(t))\|_{\rho_*, p; \mathcal{A}_0} \right) \leq C.$$

Here $\mathbf{e} = (y - y^0, w)$*, the relative velocity is* $W = V - \varepsilon\Omega X = \mathsf{E}(u, \omega)$*, and* $z = \omega + H^{-1}B_{fs}u$*. The geometry remains in a fixed small nondegenerate analytic ball.*

4.1 The relative velocity and the fast Schur variable

The constraint is $TW = 0$. Differentiate the force equation and use the identities

$$K\Omega = \Omega K, \quad M\Omega = \Omega M, \quad T'[\Omega X]^*\kappa = \Omega T^*\kappa.$$

Since $(K - H_\kappa)\Omega X = -\Omega M V$, the explicit rotation terms cancel in the full vector equation. Exactly,

$$W_t = -QM^{-1}(K - H_\kappa)W + Q_t W. \tag{4.1}$$

The normal part $Q_t W$ is fixed by differentiating $QW = W$.

Set $Z = V - \beta e$. The original force equation and (3.16) give

$$\kappa = \widehat{\kappa} + \lambda_Z, \qquad \lambda_Z = R_N^* M Z = (l_b, \delta l_c), \qquad \|l_b\|_{H^3} + \|l_c\|_{H^2} \leq C_{\text{low}}\,\|Z\|_{\mathcal{A}_0}\,.$$

This is a velocity scale, distinct from the force scale $(k_b, \lambda/\delta)$. With $G = \mathsf{E}^* M \mathsf{E}$ and $\mathbf{a} = (u, \omega)$, (4.1) becomes

$$G\mathbf{a}_t + B\mathbf{a} = \beta G\mathbf{a} + \mathsf{E}^* H_{\lambda_Z} \mathsf{E}\mathbf{a} - \mathsf{E}^* M \mathsf{E}'[V]\mathbf{a}.$$

Equivalently, $\mathbf{a}_t + G^{-1}B\mathbf{a} = \mathcal{N}_{\mathbf{a}}$, where the exact source is

$$\mathcal{N}_{\mathbf{a}} = L_{\mathrm{ref}} \left[\beta W + QM^{-1}H_{\lambda_Z} W + (R_N - R_M)T'[V]W\right], \quad R_M = M^{-1}T^*(TM^{-1}T^*)^{-1}. \tag{4.2}$$

The moment table (3.14), the velocity multiplier scale, and (3.11) give

$$\|\mathcal{N}_{\mathbf{a}}\|_{\mathcal{A}_0} \le P(\text{low norms}) \left(1 + \|\mathbf{e}\|_{\mathcal{X}_1} + \|\mathbf{a}\|_{\mathcal{A}_1}\right). \tag{4.3}$$

In each product only one factor has the higher norm. The mean source has one macro and zero optical-coordinate derivatives; the optical source has two macro and one optical-coordinate derivatives. These counts follow before estimation from Laurent division of (3.14). Coefficient derivatives of the graph use $D^2 y, Dw$; their one-higher applications give the single high geometry factor in (4.3).

The physical mass retains its optical factor:

$$G = \begin{pmatrix} G_{ss} & \delta^2 M_{sf} \\ \delta^2 M_{fs} & \delta^2 M_{ff} \end{pmatrix}.$$

Introduce the actual global operators

$$\begin{aligned} C &= H^{-1}B_{fs}, & z &= \omega + Cu, \\ \overline{M} &= G_{ss} - \delta^2 M_{sf} M_{ff}^{-1} M_{fs}, & S &= B_{ss} - B_{sf}H^{-1}B_{fs}, \\ L &= B_{sf} - M_{sf}M_{ff}^{-1}H, & N &= M_{ff}^{-1}M_{fs} - C, \qquad A_f = M_{ff}^{-1}H. \end{aligned} \tag{4.4}$$

Direct elimination of the mass equations, in this order, gives

$$\begin{aligned} \overline{M}u_t + Su + Lz &= \overline{M}\mathcal{N}_u, \\ z_t + \delta^{-2}A_f z - N\overline{M}^{-1}(Su + Lz) &= \mathcal{N}_z, \\ \mathcal{N}_z &= \mathcal{N}_\omega + C\mathcal{N}_u + C_t u. \end{aligned} \tag{4.5}$$

The source in the original mass equation was $G\mathcal{N}_{\mathbf{a}}$; its mass factors cancel in this elimination. No independent source is multiplied by δ^{-2}. Here

$$C_t = H^{-1}(B'_{fs}[V] - H'[V]C), \qquad \widehat{\kappa}'[V] = R_N^* \widehat{B} V.$$

Thus $C_t u$ has the same one-high-factor estimate as (4.3), without κ_t. The change between (u, ω) and (u, z) is bounded on $\mathcal{A}_0$; its one-higher application is tame with one $\mathcal{X}_1$ factor.

4.2 A connection for the weak macro operator

Using (3.19), define

$$\begin{aligned} a_0 &= A - C_f H^{-1} C_s, & r_0 &= b_R - C_f H^{-1} d_s, \\ l_0 &= b_L - d_f H^{-1} C_s, & v_0 &= c - d_f H^{-1} d_s. \end{aligned}$$

For arbitrary ξ, test the global matrix in Lemma 3.1 on $(\xi, -H^{-1}C_s\xi)$. Its lower row is zero, so

$$\operatorname{Re}\langle \xi, a_0 \xi\rangle_0 = \operatorname{Re}\left\langle (\xi, -H^{-1}C_s\xi), \mathbb{A}(\xi, -H^{-1}C_s\xi)\right\rangle_0 \ge c\,\|\xi\|_0^2.$$

The order-zero inverse a_0^{-1} is consequently a genuine localized global inverse. Put

$$\Gamma = a_0^{-1} r_0, \qquad \nabla = D + \Gamma. \tag{4.6}$$

Keeping all products in their order yields the exact identity

$$S = \nabla^* a_0 \nabla + (l_0 - \Gamma^* a_0)\nabla + (v_0 - l_0\Gamma). \tag{4.7}$$

The principal coefficient a_0 has H^3; Γ and the two lower coefficients have H^2 from geometry $\mathcal{X}_0$. Their H^3 norms cost one $\mathcal{X}_1$ factor.

The choice of connection matters. For the scalar graph $(1, g)$, the weak coefficients are $1 + g^2$, gg_x, gg_x, g_x^2. It gives $\Gamma = gg_x/(1 + g^2)$ and

$$S = \nabla^*(1 + g^2)\nabla + \frac{g_x^2}{1 + g^2}.$$

The strong second derivative of the graph lies in the principal connection, rather than in a lower potential.

For an operator coefficient, let $A_l(k)$ map input Fourier frequency k to $k + l$, and define

$$W_p(t)^2 = \sum_{j\geq 0} \frac{(j+1)^{2p} t^{2j}}{(j!)^2}, \qquad \|A\|^2_{\rho,p;H^r,\mathrm{coeff}} = \sum_l (1 + |\lambda_l|^2)^r W_p(\rho|\lambda_l|)^2 \sup_k \|A_l(k)\|^2 .$$

This is the coefficient analytic H^r norm in the next lemma.

Lemma 4.2. *If the actual connection has bounded coefficient analytic H^2 norm at radius ρ, its common covariant jets satisfy*

$$\sum_{j\geq 0} \varpi_j^2 \sum_{\ell=0}^{3} \left\| \nabla^{j+\ell} u \right\|_0^2 \asymp \|u\|^2_{\rho,p;H^3} .$$

The same assertion through offset 2 holds for the fast variable. Both sides use the same radius. At exponent $p + 1/2$ the comparison is tame: its high geometry dependence is linear, and its low constants depend only on exponent-p norms.

Proof. The hypothesis is precisely a bound for $\|\Gamma\|_{\rho,p;H^2,\mathrm{coeff}}$. For a circle-dependent coefficient $A(\vartheta)$ use

$$\|A\|^2_{r,p} = \sum_l (1 + |\lambda_l|^2)^r \sup_k \|A_l(k, \vartheta)\|^2_{H^p_\vartheta} .$$

Here the circle norm uses Fourier weight $(1 + |m|)^{2p}$. For $r, p > 1/2$ this is an algebra: product coefficients are $\sum_m A_{l-m}(k + m)B_m(k)$, and the two weighted convolution inequalities apply in their stated order.

For $\zeta = \rho' e^{i\vartheta}$, $0 \leq \rho' \leq \rho$, set

$$\Gamma_\zeta = e^{\zeta D}\Gamma e^{-\zeta D}, \qquad \mathcal{H}(\zeta) = e^{\zeta\nabla} e^{-\zeta D}.$$

The exact modular identity $\lambda_{k+l} - \lambda_k = e^{iks}\lambda_l$ shows that the circle norm of the transported coefficient has precisely the factor $W_p(\rho'|\lambda_l|)$; the input-frequency phase just rotates the circle. The ordered equations

$$\mathcal{H}' = \mathcal{H}\Gamma_\zeta, \qquad (\mathcal{H}^{-1})' = -\Gamma_\zeta \mathcal{H}^{-1}$$

give bounds in the same $r = 2$ algebra by their convergent iterated-integral series. Differentiating the interaction product the other way gives

$$[D, \mathcal{H}] = \mathcal{H}\Gamma_\zeta - \Gamma\mathcal{H}.$$

Its l, k coefficient contains $e^{iks}\lambda_l$. Thus $\mathcal{H}, \mathcal{H}^{-1}$ gain one coefficient derivative and belong to the $r = 3$ algebra. They act boundedly on circle H^p with values in row H^3. Circle Parseval in $e^{\zeta\nabla}u = \mathcal{H}e^{\zeta D}u$ proves same-radius comparison for jets with an ordinary H^3 base.

For each vector v, the finite norm $\sum_{\ell=0}^{3} \left\|\nabla^{\ell} v\right\|_0^2$ is equivalent to $\|v\|_{H^3}^2$. Expanding $\nabla^3 - D^3$, the highest coefficient derivative is the second derivative of Γ. It is row L^2 and acts on an H^1 field by the uniform Fourier convolution bound. Hence $\left\|(\nabla^3 - D^3)v\right\|_0 \le C\,\|v\|_{H^2}$, and triangular induction proves the finite comparison. Apply it to $v = \nabla^j u$. The offset-2 proof is the same finite expansion.

For $q = p+1/2$, the algebra has the tame bound $\|AB\|_{r,q} \le C(\|A\|_{r,q}\,\|B\|_{r,p} + \|A\|_{r,p}\,\|B\|_{r,q})$. The interaction ODE and its derivative-gain identity therefore have linear high-norm bounds, with low constants fixed at p. Since Γ is an analytic coefficient map using $D^2 y, Dw$, its high norm is bounded by $C_{\mathrm{low}}(1 + \|\mathsf{e}\|_{\rho,q;\mathcal{X}_0})$. This proves the asserted tame comparison. All series are finite-matrix entire functions before estimation, so the coefficient integrations and Parseval passages are justified by absolute convergence. □

4.3 Common jets and the energy estimate

The macro and fast coordinates both have dimension two. Apply the same operator ∇ to both, using their fixed reference coordinate identification. Put $u_m = \nabla^m u$, $z_m = \nabla^m z$, and $\mathcal{T}_m = [\partial_t, \nabla^m]$. Commuting the mass equation before substituting u_t gives exactly

$$\begin{aligned} \overline{M} u_{m,t} + S u_m + L z_m &= R_{u,m}, \\ z_{m,t} + \delta^{-2} A_f z_m - N\overline{M}^{-1}(S u_m + L z_m) &= R_{z,m}, \end{aligned} \tag{4.8}$$

where

$$\begin{aligned} R_{u,m} &= -[\nabla^m, \overline{M}]u_t - [\nabla^m, S]u - [\nabla^m, L]z + \nabla^m(\overline{M}\mathcal{N}_u) + \overline{M}\mathcal{T}_m u, \\ R_{z,m} &= -\delta^{-2}[\nabla^m, A_f]z + [\nabla^m, N\overline{M}^{-1}S]u + [\nabla^m, N\overline{M}^{-1}L]z + \nabla^m \mathcal{N}_z + \mathcal{T}_m z. \end{aligned} \tag{4.9}$$

The time insertion is $\mathcal{T}_m = \sum_{k=0}^{m-1} \nabla^k \Gamma_t \nabla^{m-1-k}$. There is no mismatch of ordinary and covariant derivatives between the two blocks.

For a coefficient operator A, the exact formulas are

$$[\nabla^m, A] = \sum_{k=1}^{m} \binom{m}{k} (\mathrm{ad}_\nabla^k A)\nabla^{m-k}, \qquad e^{\zeta\nabla} A e^{-\zeta\nabla} = \mathcal{H}(\zeta) e^{\zeta D} A e^{-\zeta D} \mathcal{H}(\zeta)^{-1}.$$

Reference positive analytic jets are $O(\rho)$; actual-minus-reference jets are $O(\eta)$ on the small analytic ball. This includes the commutator with Γ, even for matrix coefficients. The principal coefficients a_0, masses, A_f, and first-order coefficients of L, N have analytic row H^3 bounds. The lower coefficients have H^2, with one $\mathcal{X}_1$ factor for an additional derivative. Differentiating the actual products defining Γ gives

$$\Gamma_t = a_0^{-1}(r_{0,t} - a_{0,t}\Gamma).$$

The identity for $\widehat{\kappa}'[V]$ above shows that Γ_t has H^1 from the velocity base $\mathcal{A}_0$, and H^2 with one $\mathcal{A}_1/\mathcal{X}_1$ factor. Low metric time derivatives are bounded by

$$g_V = \|V_b\|_{W^{2,\infty}} + \|V_o/\delta\|_{W^{1,\infty}},$$

which is controlled by $\mathcal{A}_0$ and $\mathcal{X}_0$, using $V = \mathsf{E}(u,\omega) + \varepsilon\Omega X$.

Choose smooth real Fourier cutoffs Π_L, Π_H with $\Pi_L^2 + \Pi_H^2 = I$, low support $|\tau - I| \le 2d_0$, and high support $|\tau - I| \ge d_0$. Their phase functions are fixed; their periodized kernels have uniformly summable distance moments, by integration by parts in the phase Fourier coefficients. For each common jet let $u_L = \Pi_L u_m$, $u_H = \Pi_H u_m$, and likewise for z. Use the positive low energy

$$E_m^L = \left\langle u_L, \overline{M} u_L \right\rangle_0 + \left\langle \nabla u_L, \overline{M}\nabla u_L \right\rangle_0 + \left\langle z_L, M_{ff} z_L \right\rangle_0,$$

and the high energy

$$E_m^H = \delta^{-2} \langle \mathbf{a}_H, G\mathbf{a}_H \rangle_0 , \qquad \mathbf{a}_H = (u_H, z_H - Cu_H). \tag{4.10}$$

Let $E_m = E_m^L + E_m^H$. The current coefficients are used in these metrics, so their equivalence constants do not depend on m:

$$E_m \asymp \|u_m\|_{H^1}^2 + \|z_m\|_0^2 , \qquad D_m := \|u_m\|_{H^2}^2 + \delta^{-2} \|z_m\|_0^2 . \tag{4.11}$$

For the high part this follows from physical mass positivity, $C : H^1 \to L^2$, and $\delta^{-1} \|u_H\|_0 \asymp \|u_H\|_{H^1}$. The macro L^2 addition supplies the low zero mode.

The principal low energy estimate follows from (4.7). In differentiating its H^1 part, $[\nabla, \nabla^*]$ is order zero with one coefficient derivative of Γ. Integration by parts leaves $\operatorname{Re} \langle \nabla^2 u_L, a_0 \nabla^2 u_L \rangle_0$. Lower terms cost a small part of H^2 plus CE_m. The mass commutator contains u_t; substitute the exact macro equation in (4.5). Its bound is

$$C \|u_L\|_{H^1} \left(\|u_L\|_{H^2} + (d_0/\delta + 1) \|z_L\|_0 \right).$$

The Lz and NSu terms jointly have bound

$$C(d_0/\delta + 1) \|u_L\|_{H^2} \|z_L\|_0 ,$$

and NLz contributes $C(d_0^2/\delta^2 + 1) \|z_L\|_0^2$. Choose d_0 small, then δ small, and absorb these in the two terms of D_m. On the high band, the coercivity of the physical form B controls $\delta^{-2} \|\mathbf{a}_H\|_{\text{mixed H}^1}^2$. Since $z_H = \omega_H + Cu_H$, this controls the high contribution to D_m.

We give the remainder estimates for (4.9), including their high mass cross terms. The identity

$$E_m^H = \delta^{-2} \langle u_H, \overline{M} u_H \rangle_0 + \langle z_H + N u_H, M_{ff}(z_H + N u_H) \rangle_0 \tag{4.12}$$

shows that macro force p and fast force q pair through

$$\delta^{-2} \langle u_H, p \rangle_0 + \left\langle z_H + N u_H, M_{ff}(q + N\overline{M}^{-1} p) \right\rangle_0 .$$

Thus the additional N-application to a macro force is retained. For a high-cutoff physical source, $\|p_H\|_0 \le C\delta \|p\|_{H^1}$; its first pairing uses the macro energy root. For force commutators, the first pairing uses H^2 dissipation. The extra N-application either has one higher coefficient, paired with the energy root, or a principal term bounded by a joint macro/fast dissipation.

Take $m = j + \ell$, $\ell = 0, 1, 2$. Finite reference coefficient derivatives in the stiff fast commutator multiply lower ℓ levels. The finite Leibniz split $\binom{j+\ell}{k} = \sum_{r=0}^{\ell} \binom{\ell}{r} \binom{j}{k-r}$ separates these from analytic j derivatives. Choose fixed positive weights $\vartheta_0 = 1, \vartheta_1, \vartheta_2$, successively small relative to the lower levels, so all finite stiff Young remainders are absorbed by their lower-level fast dissipations. Positive analytic coefficient jets contribute $C(\rho + \eta)$ times the weighted dissipation. For example, the apparent factor j in a first principal commutator is canceled by $\varpi_j / \varpi_{j-1} = O(\rho/j)$ against the lower dissipative jet.

For the other commutators the relevant estimates are as follows. In $[\nabla^m, S]u$, use

$$[\nabla^m, S] = [\nabla^m, \nabla^*] a_0 \nabla + \nabla^* [\nabla^m, a_0] \nabla + [\nabla^m, l_0 - \Gamma^* a_0] \nabla + [\nabla^m, v_0 - l_0 \Gamma].$$

At $\ell = 2$ the highest force coefficient is $D^{j+4} y, D^{j+3} w$, the same-level $\mathcal{X}_0$ geometry. Its L^2-force pairing with H^2 leaves that geometry energy after Young's inequality. The principal positive jets have already been absorbed. For $[\nabla^m, \overline{M}] u_t$, first insert $u_t = -\overline{M}^{-1}(Su + Lz) + \mathcal{N}_u$; its principal orders are macro second and fast first, with the same joint low-band estimate. The remaining mass and L coefficients have the same geometry count.

Expand the fast $[\nabla^m, N\overline{M}^{-1}S]$ into its three product commutators. Losing one principal derivative leaves at most macro H^2 at the differentiated unknown. If N differentiates a high coefficient of S, the maximum is $D^{j+5}y, D^{j+4}w$, one $\mathcal{X}_1$ factor. It is paired with the fast energy root. The same bound applies to $N\overline{M}^{-1}R_{u,m}$ in (4.12). The uncommuted third-order NSu_m has already been estimated jointly in the main system. For $[\nabla^m, N\overline{M}^{-1}L]z$, the remaining first fast derivative uses the row bound $\|D\| \le (2\delta)^{-1}$; its principal part is absorbed by fast dissipation. Higher coefficient terms again have one high geometry factor.

The sole explicit stiff commutator is $-\delta^{-2}[\nabla^m, A_f]z$. Its analytic jets are small, and its finite reference jets are handled by the ϑ_ℓ choice. In the high metric its cross pairing uses

$$\|Nu_H\|_0 \le C\delta \|u_H\|_{H^2} + C \|u_H\|_0 ,$$

so small analytic jets also absorb the joint macro/fast terms. The time insertions and $\mathcal{N}_u, \mathcal{N}_z$ have one $\mathcal{A}_1$ or $\mathcal{X}_1$ factor, by (4.2)–(4.3) and the exact formulas for C_t, Γ_t. They retain their energy-root pairing.

Cutoffs are applied after common jets. For every localized coefficient A, the exact kernel difference gives

$$\|[\Pi, A]\|_{0\to 0} \le C\delta \|D_{\mathrm{coeff}} A\|_{\mathrm{kernel}} .$$

Apply this identity before expanding weak derivatives. Highest macro and mass cutoff terms have an absorbable δ; lower terms use the H^2 coefficient bound with an H^1 field, or reduce δD^2 to the available first derivative by $\delta \|D\| \le 1/2$. Mass cutoffs still use the exact substitution for u_t. In particular the stiff high cutoff term is exactly, up to sign and the real factor,

$$\delta^{-2} \langle z_H + Nu_H, M_{ff}[\Pi_H, A_f]z \rangle_0 .$$

Its absolute value is at most

$$C\delta^{-1}(\|z_H\|_0 + \|u_H\|_{H^1}) \|z\|_0 \le c_1\delta^{-2} \|z\|_0^2 + C_{c_1} E_m.$$

Choose c_1 as a small part of the global fast dissipation. Base metric derivatives, including C_t, Γ_t, use g_V and cost only $P(\text{low norms})E_m$. This accounts for all mass, principal, time, source and cutoff terms in (4.9).

It remains to sum the one-higher sources without squaring a higher norm. For a nonnegative sequence $A = (A_j)_{j\ge 0}$, let

$$\|A\|_q^2 = \sum_{j\ge 0} \frac{(j+1)^{2q}\rho^{2j}}{(j!)^2} A_j^2,$$

and define the norms of B, C in the same way. Consider

$$\sum_{k,l\ge 0} \varpi_{k+l}^2 \binom{k+l}{k} A_{k+l} B_k C_{l+1}.$$

Put $j = k + m - 1$, $m = l + 1$, and absorb weights into the sequences. The remaining factor is exactly

$$\frac{m}{\rho} \frac{(k+m)^p}{(k+1)^p (m+1)^p}.$$

Since $m \le \sqrt{j+1}\sqrt{m+1}$, weighted convolution gives

$$\sum_{k,l\ge 0} \varpi_{k+l}^2 \binom{k+l}{k} A_{k+l} B_k C_{l+1} \le \frac{C_p}{\rho} \|A\|_{p+1/2} \|B\|_p \|C\|_{p+1/2} . \tag{4.13}$$

Indeed the residual polynomial kernel is bounded by $C_p((k+1)^{-p} + (m+1)^{-p})$, and each reciprocal sequence is square summable. Repeated ordered products and inverse series retain

one differentiated factor. Lemma 4.2 at half exponent converts its covariant and ordinary norms tamely. After squaring that tame norm comparison the extra half-weight velocity norm is bounded by a low constant times the sum of the connection half-weight velocity energy and the ordinary half-weight geometry energy.

Set

$$E_x = \|\mathbf{e}\|^2_{\rho,p;\mathcal{X}_0}\,, \qquad E_v = \sum_{j\ge 0} \varpi_j^2 \sum_{\ell=0}^{2} \vartheta_\ell E_{j+\ell},$$

$$D_v = \sum_{j\ge 0} \varpi_j^2 \sum_{\ell=0}^{2} \vartheta_\ell D_{j+\ell}.$$

Let $\mathcal{L}_x, \mathcal{L}_v$ be the respective energy sums with an extra factor j. In particular E_v is equivalent to the ordinary analytic $\mathcal{A}_0$ norm. In scaled mean/optical components the geometry equation is

$$(y_t, w_t) = \left((\mathsf{E}(u, z - Cu))_b, \frac{(\mathsf{E}(u, z - Cu))_o}{\delta} \right) + \varepsilon(\Omega y, \Omega w).$$

Equation (3.11) has one higher factor, so (4.13) applies to it too. The $j = 0$ part of the half weights is ordinary growth, with a fixed constant depending on a positive lower radius.

Combining the estimates, choose ρ, η small to absorb all principal $C(\rho + \eta)D_v$ terms. For a fixed geometry weight M_w, the energy $\mathcal{E} = M_w E_x + E_v$ satisfies

$$\mathcal{E}_t + cD_v \le P_{M_w}(\mathcal{E})(1 + \mathcal{E}) + \left(\frac{2\rho'}{\rho} + \frac{P_{M_w}(\mathcal{E})}{\rho} \right) (M_w \mathcal{L}_x + \mathcal{L}_v). \tag{4.14}$$

The polynomial has fixed reference and positive-lower-radius constants; none depends on the mesh.

4.4 Continuation

Choose the compact geometric neighborhood and β, then the finite ϑ_ℓ weights and d_0. Choose η, ρ_0 small for the preceding absorptions. Initial consistency and reference analyticity give $E_x(0) = 0$, $E_v(0) \le C_0$, including the bounded scaled optical initial layer. Choose $M_w > (2C_0 + 1)/\eta^2$, and fix $\rho_* = \rho_0/2$. Let $\nu > P_{M_w}(2C_0 + 1)$, put $\rho(t) = \rho_0 - \nu t$, and choose $T_0 > 0$ smaller than $\rho_0/(2\nu)$ and the scalar comparison time for $\mathcal{E} \le 2C_0 + 1$. Then (4.14) implies this energy bound and, strictly, $E_x < \eta^2$. This supplies the same small analytic geometry neighborhood used in the coefficient estimates.

For every fixed mesh, all factorial jet sums and their differentiated sums converge absolutely on compact local time intervals, since D, ∇ and their time derivatives are finite matrices. If the original maximal time were below T_0, bounded physical velocity would give a limit for X. The scaled edge cross products and the full weak-normal block retain strict positive gaps at that limit. The local ODE therefore extends by Picard iteration, a contradiction. The cyclic reduction remains the full original solution by uniqueness. This proves Proposition 4.1.

5 An optical label and the map error

5.1 The label and its spatial equation

At row i, let $p_i = Dy_i$, $\hat{e}_i = p_i/|p_i|$, and define

$$q_i = \Omega y_i - (\Omega y_i \cdot \hat{e}_i)\hat{e}_i, \qquad A_i = |q_i| > 0, \qquad \hat{t}_i = q_i/A_i, \qquad \hat{n}_i = \hat{t}_i \times \hat{e}_i.$$

These quantities are smooth on the small neighborhood already obtained. Put

$$k_i = \frac{w_i \cdot \hat{t}_i}{A_i}, \qquad F_i = \frac{w_i \cdot \hat{n}_i}{A_i}.$$

For k, F near zero, define

$$d_a = (a-2k, -2F), \quad d_b = (b+2k, 2F), \quad L_a = |d_a|, \quad L_b = |d_b|, \quad U = L_a+L_b, \quad V_c = L_b-L_a$$

and the scalar label

$$I(k,F) = F\left[\frac{256}{27(U+1)^3(1-V_c)(1+V_c)^2}\right]^{1/3}. \tag{5.1}$$

The positive cube root is smooth near zero and $I_F(0,0) = 1$. Let $\mathcal{J}(x,y) = (-y,x)$ in the $(\hat{t},\hat{n})$ plane. The coherent extruded-cell normals are $N_a = \mathcal{J}d_a/L_a$, $N_b = \mathcal{J}d_b/L_b$. Their weighted sum is $\bar{n} = aN_a + bN_b = \mathcal{J}\mathcal{T}$, where $\mathcal{T} = ad_a/L_a + bd_b/L_b$.

Direct differentiation in (k,F) verifies the label identity. The local gradients of U, V_c are orthogonal, with

$$|\nabla_{k,F}U|^2 = \frac{16(U^2-1)}{U^2-V_c^2}, \qquad |\nabla_{k,F}V_c|^2 = \frac{16(1-V_c^2)}{U^2-V_c^2},$$
$$16F^2 = (U^2-1)(1-V_c^2).$$

Moreover $\mathcal{T} = ((b-a)\nabla_{k,F}U + \nabla_{k,F}V_c)/4$. For $F \neq 0$, (5.1) gives

$$\partial_U \log|I| = \frac{1}{U^2-1}, \qquad \partial_{V_c}\log|I| = -\frac{1}{3(1-V_c^2)}.$$

Since $b-a = 1/3$, $\nabla_{k,F}I \cdot \mathcal{T} = 0$. Continuity extends this across $F = 0$. As $\mathcal{T}(0,0) = (1,0)$, there is a smooth bounded $\mu > 0$, with bounded reciprocal, such that

$$\nabla_{k,F}I = \mu\bar{n}, \qquad \partial_w I_i = (\mu_i/A_i)\bar{n}_i. \tag{5.2}$$

An optical component parallel to p_i has no component in this identity, since it does not change the plane normal.

We now use the actual curved constraint, whose matrix T still contains the original geometric normals. Write $W_\pm = \xi \pm \delta\zeta$. Its optical row is exactly

$$T_{ob}\xi + \delta T_{oo}\zeta = 0.$$

Whole-rectangle integration and exact divided remainders give

$$\sum_j T_{ob}(i,j) = \delta b_i, \qquad \sum_j T_{oo}(i,j) = m_i\bar{n}_i + \delta c_i, \qquad m_i > 0.$$

Here m_i is a reference scalar weight bounded away from zero. Subtract neighboring ξ_i, ζ_i values from the finite-band row sums, and divide by the common δ. Every difference is $s = 4\delta$ times a finite sum of forward differences. Thus

$$\bar{n}_i \cdot \zeta_i = B_i\xi_i + \sum_j C_{ij}D\xi_j + \delta\sum_j E_{ij}D\zeta_j + \delta G_i\zeta_i. \tag{5.3}$$

The coefficients here are exactly the finite combinations from $T_{ob}, T_{oo}, b_i, c_i, m_i^{-1}$; they have uniform bounds and Lipschitz dependence at geometry $\mathcal{X}_0$. Their lower descriptors are D^2y, Dw.

The position equation reads

$$y_t = \xi + \varepsilon\Omega y, \qquad w_t = \zeta + \varepsilon\Omega w, \qquad p_t = D\xi + \varepsilon\Omega p.$$

The scalar $I(y,p,w)$ is invariant under simultaneous rotation. The explicit ε terms therefore cancel in its chain rule. Inserting (5.3) into

$$I_{i,t} = \partial_y I_i \cdot \xi_i + \partial_p I_i \cdot D\xi_i + (\mu_i/A_i)\bar{n}_i \cdot \zeta_i$$

gives an exact equation of the form

$$I_t = \mathcal{F}_\delta(\mathbf{e}, \xi, D\xi) + \delta \mathcal{G}_\delta(\mathbf{e}, \zeta, D\zeta). \tag{5.4}$$

On the uniform interval, $\mathcal{F}_\delta$ is uniformly locally Lipschitz from $\mathcal{X}_0 \times H^1$ to L^2, and the second term has uniformly bounded coefficient functions. This equation retains the spatial and moving-frame terms.

5.2 A fixed positive-time effect

The positive lower analytic radius supplies an extra fixed row derivative with constant C/ρ_*. Proposition 4.1 and (3.11) consequently give

$$\|\mathbf{e}_t\|_{\mathcal{X}_0} \le C, \qquad \|\mathbf{e}(t)\|_{\mathcal{X}_0} \le Ct.$$

The macro equation in (4.5) gives $\|u_t\|_{H^1} \le C$ without a time derivative of z. The physical mean relative velocity is

$$\xi = \mathsf{E}_{bb}(\mathbf{e})u + \delta^2 \mathsf{E}_{bo,1}(\mathbf{e})\omega, \qquad \omega = z - Cu,$$

where $\mathsf{E}_{bo,1}$ has one row order. The second term has size $O(\delta^2)$ in H^1, since ω is bounded in H^2. Bounding its value at both times, rather than differentiating it through the fast layer, gives

$$\|\xi(t) - \xi(0)\|_{H^1} \le C(t + \delta^2).$$

Equation (5.4) then implies

$$\|I_t(t) - I_t(0)\|_0 \le C(t + \delta). \tag{5.5}$$

It remains to calculate the initial rate. At the reference, $I_y = I_p = 0$ and $I_w = n_0/A + O(\delta)$. Each short or long reference quadrilateral is exactly planar. After rotating its azimuthal midpoint to zero, its vertices are

$$(\rho_i \cos(d/2),\ \pm\rho_i \sin(d/2), z_i), \qquad (\rho_{i+1} \cos(d/2),\ \pm\rho_{i+1} \sin(d/2), z_{i+1}).$$

Its two geometric normals agree. The trace of $\varepsilon\Omega Y$ on that normal has leading magnitude $\varepsilon A \cos\theta \sin(d/2)$, with opposite species signs. The short strip has the χ sign and the long strip its negative. For a lower optical row test on the two leading triangles, the consistent mass gives $A_{\mathrm{tri}}/6 + A_{\mathrm{tri}}/6 = A_{\mathrm{tri}}/3$; the upper row gives the same pairing with the global χ. After neighboring-patch assembly, the reference area factors cancel between the optical mass and forcing. In the equation for $V_c = V_o/\delta$, the signed forcing-to-mass quotient is

$$\frac{a\sin(a\delta/2) - b\sin(b\delta/2)}{\delta(a+b)} = \frac{a^2 - b^2}{2(a+b)} + O(\delta^2).$$

The common triangle factor $A_{\mathrm{tri}}/3$ cancels; the short-strip term has positive sign and the long-strip term negative sign. The remaining curved-row variations contribute $O(\delta)$. Therefore

$$n_0 \cdot V_c(0) = \varepsilon A \cos\theta \, \frac{a^2 - b^2}{2(a+b)} + O(\delta) = -\frac{\varepsilon A \cos\theta}{6} + O(\delta). \tag{5.6}$$

Here $V_c = V_o/\delta$. The term $T_{ob}V_b(0)$, after division of the optical equation by δ, is $O(\delta)$, since T_{ob} is δ times first row order and $V_b(0) = O(\delta)$ in H^1. Thus $I_t(0) = -\varepsilon\cos\theta/6 + O(\delta)$ in row L^2.

Since $I(0) = 0$, integration of (5.5) yields

$$\|I(t) + \varepsilon t\cos\theta/6\|_0 \le C(t^2 + \delta t). \tag{5.7}$$

For sufficiently large n, the exact uniform-grid norm is $\|\cos\theta\|_0 = \sqrt{\pi}$. Fix one

$$0 < t_* < \min\{T_0, \varepsilon\sqrt{\pi}/(24C)\},$$

and then take δ small. It follows that

$$\|I(t_*)\|_0 \ge \varepsilon t_*\sqrt{\pi}/12 > 0. \tag{5.8}$$

The time is independent of the mesh and ε has already been fixed.

5.3 The original energy and the lifted error

On the small neighborhood, $|I_i| \le C|w_i|$. For $E_h = X_h - Y_h$, the leading short and long azimuthal edge differences in its aligned representation are

$$\delta\{a\Omega(y - y^0) - 2w\}, \qquad \delta\{b\Omega(y - y^0) + 2w\}.$$

Their two squared norms control w, including every Cartesian component. The exact rotation formulas differ by uniformly small terms, absorbed with the physical L^2 part. Reference shape regularity therefore gives from (5.8)

$$\|E_h(t_*)\|^2_{H^1(\Gamma_h^0)} \ge c\varepsilon^2 t_*^2. \tag{5.9}$$

Test the original force equation by V and ΩX. The weak constraint gives $TV = \varepsilon T\Omega X$, while $\langle KX, \Omega X\rangle_0 = 0$ by componentwise skew rotation. Hence the exact identity is

$$\frac{1}{2}\frac{d}{dt}\langle X, KX\rangle_0 + \|V\|_M^2 = \varepsilon\langle V, M\Omega X\rangle_0. \tag{5.10}$$

Here the normalized row pairings equal the full reference surface integrals on cyclic fields, and $\|V\|_M^2 = \langle V, MV\rangle_0$.

Put $d_h = \sup_{0\le t\le t_*}\|E_h(t)\|_M$. Expand $X = Y + E_h$ in (5.10). The work integral is

$$\varepsilon\langle E_h(t), M\Omega Y\rangle_0 + \varepsilon\int_0^t \langle V, M\Omega E_h\rangle_0\, d\tau.$$

The reference bound (3.15) controls $\langle E_h, KY\rangle_0$. Proposition 4.1 bounds physical V uniformly in L^2. Thus

$$\frac{1}{2}\left\|\nabla_{\Gamma_h^0} E_h(t)\right\|_{L^2}^2 + \int_0^t \|V\|_M^2\, d\tau \le Cd_h. \tag{5.11}$$

Combining (5.9) with this bound gives fixed positive constants α, C_1, C_2, D_0, independent of the mesh, such that

$$\alpha := c\varepsilon^2 t_*^2 \le C_1 d_h + C_2 d_h^2, \qquad d_h \le D_0.$$

Consequently $d_h \ge \alpha/(C_1 + C_2D_0) > 0$. Lemma 2.1 gives a fixed lift norm constant $c_\ell > 0$ and an $O(\delta^2)$ identity error, so

$$\sup_{t\le t_*}\left\|X_h^\ell(t) - \mathrm{id}\right\|_{L^2(\Gamma^0)} \ge \frac{c_\ell\alpha}{C_1 + C_2D_0} - C\delta^2.$$

For sufficiently small δ, take $T_* = t_*$ and $c_* = c_\ell\alpha/[2(C_1 + C_2D_0)]$. These are positive fixed constants. The continuous solution is the identity, the closest lift is the original one, and the full original discrete solution exists nondegenerately on this interval. This proves Theorem 2.2.

6 Convergence for cubic elements

We now consider arbitrary smooth data, rather than the stationary torus. Write $u = U_{\mathrm{ext}}$ in this section. Assume that (2.1) has a smooth solution on a fixed finite interval $[0,T]$, with smooth flow map and inverse throughout that interval. Let Γ_h^0 be any admissible shape-regular, quasi-uniform cubic nodal interpolant of Γ^0. For its element parametrizations require exactly the original bounds

$$\max_K \big(\|F_K\|_{W^{3,\infty}(K_f)} + \|F_K^{-1}\|_{W^{1,\infty}(K)}\big) \le C_0. \tag{6.1}$$

Here K_f is the flat triangle with the same vertices as K. The space S_h now consists of continuous functions whose pullbacks by F_K are cubic polynomials. We retain precisely (2.2)–(2.4), including their fixed reference measure and discontinuous current unit normals.

Theorem 6.1. *For every such smooth solution and mesh family there are $h_0, C_T > 0$, independent of h, such that for $h < h_0$ the original semidiscrete solution exists uniquely on $[0,T]$. Its map remains an embedding and its element differentials remain nondegenerate. Moreover,*

$$\sup_{0\le t\le T} \big\|X_h^\ell(t) - X(t)\big\|_{L^2(\Gamma^0)} \le C_T h^3 \sqrt{1+|\log h|}. \tag{6.2}$$

If $d(x,t)$ is signed distance to $\Gamma(t)$, then also

$$\sup_{0\le t\le T} \|d(X_h(\cdot,t),t)\|_{L^2(\Gamma_h^0)} \le C_T h^3.$$

The constants may depend on the fixed smooth solution, T, and the uniform mesh-family bounds.

The proof controls shape error before estimating the full map error. The first energy is a relative volume weighted by signed distance. The second is the energy of a weakly tangent part of the map error. Their interaction leaves a vanishing coefficient in the shape equation, while allowing a logarithmic loss in the tangent estimate.

6.1 Reference geometry and weak tangency

All notation in this section is local; in particular D below is a gradient norm, rather than the row difference of the preceding sections. Put

$$(w,z)_h = \int_{\Gamma_h^0} w\cdot z, \qquad a_h(w,z) = \int_{\Gamma_h^0} \nabla w : \nabla z, \qquad (L_h w, z)_h = a_h(w,z).$$

Unqualified norms and gradients use this fixed surface. Let Π_h and Π_h^3 be the scalar and vector mass projections. With the initial closest-point projection $a : \Gamma_h^0 \to \Gamma^0$, define

$$\overline{X} = X\circ a, \quad Y = I_h \overline{X}, \quad Z_h = Y_t, \quad m = n(\overline{X},t), \quad \nu = N(Y), \quad \kappa^* = I_h(\kappa\circ a).$$

The known unit field m is the restriction of a smooth field on a fixed initial tube. It is used only in the analysis.

The original mesh assumptions imply

$$\|Y - \overline{X}\|_\infty \le Ch^4, \qquad \|\nabla(Y-\overline{X})\|_\infty + \|\nu - m\|_\infty \le Ch^3. \tag{6.3}$$

Indeed, derivatives of F_K through order three are bounded and its fourth derivatives vanish. Composing any fixed smooth ambient function with F_K therefore gives bounded fourth derivatives. Subtract its cubic Taylor polynomial and use the fixed nodal basis on the scaled triangle. This gives interpolation errors Ch^4 and Ch^3 in value and gradient. Applied to the

initial signed distance, which vanishes at all nodes, it gives initial distance Ch^4 and initial normal error Ch^3. The closest-point area ratio is $1 + O(h^4)$, and the pulled-back Dirichlet metric differs by $O(h^4)$: its displacement contribution is $O(h^4)$, and its squared normal-slope contribution is $O(h^6)$.

Adjacent elements have the same polynomial edge tangent. Their outward conormals therefore have sum $O(h^3)$. For a known smooth ambient coefficient b, this gives

$$[\partial_\mu b] = O(h^3). \tag{6.4}$$

Scaled polynomial norm equivalence also gives

$$\begin{aligned} \|w\|_\infty &\le Ch^{-1} \|w\|_2 , & \|\nabla w\|_2 &\le Ch^{-1} \|w\|_2 , \\ \|\nabla w\|_\infty &\le Ch^{-2} \|w\|_2 , & \|w\|_{L^2(\text{edges})} &\le Ch^{-1/2} \|w\|_2 . \end{aligned} \tag{6.5}$$

The last estimate is summed over element traces, with bounded overlap. In particular $\|L_h w\|_2 \le Ch^{-1} \|\nabla w\|_2$. These geometric estimates concern the known initial mesh, not the curvature of the unknown evolving surface.

For a smooth coefficient b and an FE function w, product superapproximation takes the form

$$\|bw - I_h(bw)\|_2 + h \|\nabla(bw - I_h(bw))\|_2 \le Ch^2 \|w\|_{H^1} . \tag{6.6}$$

To see the two powers, remove a constant value of b, whose product with w is reproduced. Subtract also the element mean of w. The latter difference has norm at most $Ch \|\nabla w\|_2$, by the line-segment formula on the flat triangle. In the fourth-order interpolation remainder, the highest remaining polynomial derivative is bounded by $Ch^{-2} \|\nabla w\|_2$, so multiplication by h^4 gives h^2. The constant part of w gives $Ch^4 \|w\|_2$. Differentiating the scaled basis once proves the gradient assertion. Thus

$$\big\|(I - \Pi_h^3)(m\chi)\big\|_2 \le Ch \|\chi\|_2 , \qquad \big\|\Pi_h^3(m\chi) - m\chi\big\|_{H^1} \le C \|\chi\|_2 . \tag{6.7}$$

For the second bound split at $I_h(m\chi)$ and use (6.5); no H^1 stability of Π_h is needed.

For any measurable unit field b, set

$$\mathcal{T}_b v = \Pi_h(b \cdot v), \qquad \mathcal{T}_b^* \chi = \Pi_h^3(b\chi), \qquad \mathcal{S}_b = \mathcal{T}_b \mathcal{T}_b^* .$$

Mass orthogonality gives the exact identity

$$(\mathcal{S}_b \chi, \chi)_h = \|\chi\|_2^2 - \big\|(I - \Pi_h^3)(b\chi)\big\|_2^2 . \tag{6.8}$$

Hence $\mathcal{S}_m \ge 1 - Ch^2$. Write

$$\mathcal{T} = \mathcal{T}_m, \quad \mathcal{S} = \mathcal{S}_m, \quad Q = I - \mathcal{T}^* \mathcal{S}^{-1} \mathcal{T}, \quad e = X_h - Y = z + \mathcal{T}^* \psi,$$

$$z = Qe, \quad \psi = \mathcal{S}^{-1} \mathcal{T} e, \qquad E = \|e\|_2 , \quad F = \|\psi\|_2 , \quad Z = \|z\|_2 , \quad D = \|\nabla z\|_2 .$$

Then Q is the mass-orthogonal projection onto $\ker \mathcal{T}$,

$$E^2 = Z^2 + (\mathcal{S}\psi, \psi)_h \asymp Z^2 + F^2, \qquad (e_t, z)_h = \tfrac{1}{2}(Z^2)' - \tfrac{1}{2}(e, Q_t e)_h, \qquad \|Q_t\|_{2\to 2} \le C. \tag{6.9}$$

The time bound follows by differentiating the inverse and using $\mathcal{T}_t v = \Pi_h(m_t \cdot v)$.

Weak tangency yields a stronger leakage estimate than an inverse bound:

$$\|m \cdot z\|_2 \le Ch^2 (D + Z), \qquad \|\nabla(m \cdot z)\|_2 \le Ch(D + Z). \tag{6.10}$$

Indeed, $\Pi_h(m\cdot z)=0$, so $I_h(m\cdot z)=\Pi_h[I_h(m\cdot z)-m\cdot z]$. Apply (6.6) and then an inverse estimate. Furthermore,

$$\begin{aligned} |a_h(z,\mathcal{T}^*\chi)|\le C(D+Z)\,\|\chi\|_2\,, \quad \|\mathcal{T}L_h z\|_2\le C(D+Z),\\ \|QL_h\mathcal{T}^*\psi\|_2\le Ch^{-1}F, \qquad a_h(e,z)\ge \tfrac12 D^2-CE^2. \end{aligned} \tag{6.11}$$

For the first assertion, replace $\mathcal{T}^*\chi$ by $m\chi$ using (6.7), and expand

$$a_h(z,m\chi)=(\nabla\chi,\nabla(m\cdot z))_h-\sum_j(\nabla\chi,z_j\nabla m_j)_h+\sum_j(\chi\nabla m_j,\nabla z_j)_h.$$

The first term uses (6.10). Integrate the middle term on each initial element; the volume coefficients are bounded, and (6.4) with two inverse traces bounds its edge term by $Ch^2\,\|\chi\|_2\,Z$. The last term is bounded directly. Adjoint testing gives the third assertion of (6.11); Young's inequality gives the last.

We also use the two-dimensional discrete logarithmic inequality

$$\|z\|_\infty\le C\sqrt{\ell}(D+Z), \qquad \ell=1+|\log h|. \tag{6.12}$$

Here is a proof uniform over these curved meshes. The element mass matrix gives a polynomial evaluation dual δ_h, with $\|\delta_h\|_1\le C$, $\|\delta_h\|_2\le Ch^{-1}$. Push its density into a smooth initial coordinate chart. With $\widehat{f}(\xi)=\int f(x)e^{-2\pi ix\cdot\xi}\,dx$, Plancherel's identity [2, Theorem 43] and $|\widehat{\delta_h}|\le C$ give

$$\int_{\mathbb{R}^2}\frac{|\widehat{\delta_h}(\xi)|^2}{1+4\pi^2|\xi|^2}\,d\xi\le C\int_{|\xi|\le h^{-1}}\frac{d\xi}{1+|\xi|^2}+Ch^2\,\|\delta_h\|_2^2\le C\ell.$$

Weighted Cauchy–Schwarz bounds its pairing with a chart H^1 function. The Fourier derivative identity follows by integration by parts and then convolution approximation. A fixed finite chart cover, with cutoffs equal to one on the relevant elements, proves (6.12).

6.2 The actual mixed system on a small-error interval

Fix the analysis exponent $1/4$, and work initially while

$$E\le h^{11/4}, \qquad \rho=C_1(h^3+h^{-2}E), \qquad \alpha=h+\rho. \tag{6.13}$$

The fixed C_1 is sufficiently large. For small h, $\rho\le Ch^{3/4}$, $\alpha\ge h$, $\alpha^2\le h$, and $\alpha^2\le Ch^{3/2}$. All constants below are uniform on one fixed small geometric neighborhood and independent of the later bulk constant.

For the actual normal $N=N(X_h)$,

$$\|N-m\|_\infty\le C\rho, \qquad \|N-m\|_2\le C(h^{-1}F+E+\alpha D+h^3). \tag{6.14}$$

The first variation about Y is $-\mathcal{H}(\mathrm{D}e)^{\mathsf{T}}\nu$, where $\mathcal{H}$ is a uniformly bounded inverse-tangent coefficient. Use $(\mathrm{D}e)^{\mathsf{T}}m=\nabla(m\cdot e)-(\mathrm{D}m)^{\mathsf{T}}e$, (6.10), and $\|\nabla e\|_2\le C(D+h^{-1}F)$. The linear term is bounded by $C(h^{-1}F+E+hD)$; its Taylor remainder is at most $C\rho(D+h^{-1}F)$. Adding the reference normal error proves (6.14).

Set $\mathcal{U}=\mathcal{T}_N$ and $Q_N=I-\mathcal{U}^*\mathcal{S}_N^{-1}\mathcal{U}$. The exact identity (6.8) gives

$$\mathcal{S}_N\ge[1-(Ch+C\rho)^2]I, \qquad \|Q_N-Q\|_{2\to2}\le C\rho. \tag{6.15}$$

The inverse-difference formula proves the second bound. Discontinuity of N across edges is permitted; no derivative of N is used.

Define the known defects

$$\mathfrak{d} = Z_h + L_h Y - \mathcal{T}_\nu^* \kappa^*, \qquad r_h = \Pi_h[(Z_h - u(Y,t)) \cdot \nu].$$

Pulling the smooth weak equation to the initial mesh and using (6.3) gives

$$\|\mathfrak{d}\|_{H_h^{-1}} \le Ch^3, \qquad \|\mathfrak{d}\|_2 \le Ch^2, \qquad \|r_h\|_2 \le Ch^3. \tag{6.16}$$

The dual norm here tests only FE vectors in their full H^1 norm. The stiffness interpolation error is Ch^3, while initial metric errors are $O(h^4)$. Inverse testing gives the mass bound. For the bottom defect compare with $\overline{X}_t - u(\overline{X})$, which is smooth and exactly m-tangent.

For $\mu = \kappa_h - \kappa^*$, subtraction gives exactly

$$e_t + L_h e = \mathcal{U}^* \mu + G, \qquad \mathcal{U} e_t = R, \tag{6.17}$$

where

$$\begin{aligned} G &= \Pi_h^3[\kappa^*(N-\nu)] - \mathfrak{d}, \\ R &= \Pi_h[(u(X_h) - u(Y)) \cdot N + (u(Y) - Z_h) \cdot (N - \nu)] - r_h. \end{aligned}$$

Thus $\|G\|_2 + \|R\|_2 \le C(h^{-1}F + E + \alpha D + h^2)$, with h^3 in the separate bound for R. Exact elimination gives

$$\mu = \mathcal{S}_N^{-1}(R + \mathcal{U} L_h e - \mathcal{U} G), \qquad e_t = -Q_N L_h e + Q_N G + \mathcal{U}^* \mathcal{S}_N^{-1} R.$$

By (6.11) and $\|\mathcal{U} - \mathcal{T}\| \le C\rho$,

$$\begin{aligned} \|\mu\|_2 &\le C(h^{-2}F + \alpha h^{-1} D + E + h^2), \\ \|e_t\|_2 &\le C(h^{-1}D + \alpha h^{-2} F + E + h^2). \end{aligned} \tag{6.18}$$

For the improved normal coefficient in the second line, keep the projection before the stiffness:

$$\|Q_N L_h \mathcal{T}^* \psi\|_2 \le C(h^{-1} + \rho h^{-2}) F.$$

Finally, the actual pointwise residual $q = (v_h - u(X_h, t)) \cdot N$ satisfies $\Pi_h q = 0$ and

$$\|q\|_2 \le C(h^{-1}F + \alpha h^{-1} D + E + \alpha h^2). \tag{6.19}$$

Indeed, its $e_t \cdot m$ projection defect costs $Ch \|e_t\|_2$, and its $e_t \cdot (N - m)$ part costs $C\rho \|e_t\|_2$. The remaining reference-tangent part costs $C(\|N - m\|_2 + E + h^4)$. Substitution of (6.18) gives an additional $\alpha^2 h^{-2} F$, absorbed by $\alpha^2 \le h$; $h^3 \le \alpha h^2$ handles the source.

6.3 Relative volume and the area-weighted projection defect

Orient one connected component outward, and let $\Omega(t)$ and $\Omega_h(t)$ be its exact and numerical bounded interiors. The small geometric neighborhood gives a normal graph $r(s,t)$ over $\Gamma(t)$. Define

$$\mathcal{B}(t) = \int_{\mathbb{R}^3} (\mathbf{1}_{\Omega_h(t)} - \mathbf{1}_{\Omega(t)}) d(x,t)\, dx. \tag{6.20}$$

Only a fixed tube is involved, so signed distance may be smoothly extended outside it. In normal coordinates,

$$\mathcal{B} = \int_{\Gamma(t)} \int_0^{r(s,t)} \xi \det(I + \xi \mathsf{S}(s,t))\, d\xi\, dS_s,$$

where S is the exact shape operator, with the sign matching this normal-coordinate Jacobian. The integral is nonnegative also when $r < 0$. Uniform graph and area bounds give, for $f = d(X_h, t)$,

$$c\,\|f\|_2^2 \le \mathcal{B} \le C\,\|f\|_2^2\,. \tag{6.21}$$

For several components take the sum of these functionals separately, so nesting causes no cancellation. Reversing a component's orientation reverses its multiplier and leaves the position equations unchanged.

Let J be the area ratio of X_h from the fixed initial mesh to the current numerical surface. Closed-surface transport gives

$$\mathcal{B}' = \int (\mathbf{1}_{\Omega_h} - \mathbf{1}_{\Omega})[d_t + \operatorname{div}(du)]\,dx + (q, fJ)_h. \tag{6.22}$$

To verify the formula for piecewise-smooth elements, differentiate their swept regions. Each face contributes its area times normal displacement; the common moving-edge traces cancel all side fluxes. Add and subtract u on the two closed boundaries and apply the divergence identity to du. The exact normal speed equals $u \cdot n$, giving (6.22). No derivative of the numerical normal appears.

On the exact surface, $d_t + u \cdot \nabla d = 0$. Integrating its smooth normal derivative along tube fibers shows that this expression is $c_1(x,t)d$ with bounded c_1. The volume term in (6.22) is therefore at most $C\mathcal{B}$. The ORIGINAL weak constraint, in the fixed measure, removes only an FE projection:

$$\mathcal{B}' \le C\mathcal{B} + (q, (I - \Pi_h)(fJ))_h. \tag{6.23}$$

Thus the current area ratio J must remain in the test defect.

The estimate that permits closure is

$$\|(I - \Pi_h)(fJ)\|_2 \le C\{(h + \rho^2)F + h\alpha(D + E) + h^4\} \le C\{hF + h\alpha(D + E) + h^4\}. \tag{6.24}$$

We give all parts of its proof. Let j be the known smooth exact flow area ratio, pulled back to the initial mesh. Distance Taylor expansion at $\overline{X}$ gives

$$f = m \cdot e + \mathcal{R}_f, \qquad \|\mathcal{R}_f\|_2 \le C(h^4 + h^{-1}E^2), \qquad \|f\|_\infty \le C(h^4 + h^{-1}E). \tag{6.25}$$

Use the weak leakage of z, $\left\|m \cdot \Pi_h^3(m\psi) - \psi\right\|_2 \le ChF$, and the smooth-product defect of $j\psi$. They imply

$$\|(I - \Pi_h)(jf)\|_2 \le C\{hF + h^2(D + E) + h^4 + h^{-1}E^2\}. \tag{6.26}$$

For a rank-two differential matrix A, write $\mathcal{J}(A) = \sqrt{\det(A^{\mathsf{T}}A)}$. Its derivative at the exact differential satisfies

$$\mathrm{D}\mathcal{J}(\mathrm{D}\overline{X})[m \otimes \xi] = 0, \tag{6.27}$$

since $(\mathrm{D}\overline{X})^{\mathsf{T}}m = 0$. Write $e = z + m\psi + r$, where $r = \Pi_h^3(m\psi) - m\psi$ has H^1 norm at most CF. The term $m\nabla\psi$ therefore drops out of the linear area variation. Its other linear terms cost $C(D + E)$, and its quadratic remainder costs $C\rho(D + h^{-1}F)$. Interpolation and initial lift errors give

$$\|J - j\|_2 \le C\{h^3 + D + E + \rho(D + h^{-1}F)\}. \tag{6.28}$$

Multiply by the last bound in (6.25). The inequality $E/h \le Ch\rho$ gives

$$\|f(J - j)\|_2 \le C\{(h + \rho^2)F + h\alpha(D + E) + h^4\}.$$

Keep the term $h^{-1}E^2$ in (6.26) until using $h^{-1}E^2 \le Ch\rho E$. Combining proves the first inequality in (6.24); $\rho^2 \le \alpha^2 \le h$ proves the second.

The same decomposition, absorbing ChF, gives

$$F \le C\,\|f\|_2 + Ch^2(D+E) + Ch^{-1}E^2 + Ch^4.$$

Since $D \le Ch^{-1}E$,

$$F^2 \le C\mathcal{B} + Ch^2E^2 + Ch^{-2}E^4 + Ch^8. \tag{6.29}$$

Pair (6.19) and (6.24) in (6.23). Their main terms are F^2, αFD, and α^2D^2. All cross terms are bounded by $C\{F^2 + \alpha^2(D^2+E^2) + h^6\}$. In particular the source h^4E costs $\alpha^2E^2 + h^8/\alpha^2 \le \alpha^2E^2 + h^6$; the other source pairings use Young's inequality in the same way. Under (6.13), $h^{-2}E^4 \le h^{7/2}E^2 \le h^2E^2$. Thus

$$\mathcal{B}' \le C\mathcal{B} + C\alpha^2(D^2+E^2) + Ch^6. \tag{6.30}$$

The vanishing coefficient α^2 permits control of shape error without first knowing an optimal bound for the whole map.

6.4 Tangent energy and the fourth power of normal error

Test (6.17) by z. Equations (6.9) and (6.11) give

$$\tfrac{1}{2}(Z^2)' + \tfrac{1}{2}D^2 \le CE^2 + (G,z)_h + (\mu,(\mathcal{U}-\mathcal{T})z)_h. \tag{6.31}$$

For any fixed $\vartheta > 0$, small h gives

$$|(G,z)_h| \le \vartheta D^2 + C_\vartheta(E^2+h^6). \tag{6.32}$$

The defect $\mathfrak{d}$ uses its H_h^{-1} bound. For the linear normal variation write

$$-\mathcal{H}\nabla(e\cdot m) + \mathcal{H}(\mathrm{D}m)^{\mathsf{T}}e - \mathcal{H}(\mathrm{D}e)^{\mathsf{T}}(\nu - m).$$

Replacing $\mathcal{H}$ by the exact-reference coefficient costs $Ch^3\,\|\nabla e\|_2\,E \le Ch^2E^2$. The replacement of κ^* by its smooth reference value is smaller. Integrate the first term once on each initial element. Volume terms cost $CE(D+E)$. The known coefficient flux jumps are $O(h^3)$, since their dependence on the initial tangent planes is smooth and their conormals have the jump in (6.4). Two inverse traces bound the edge terms by CE^2.

For the nonlinear remainder, choose an orthonormal reference tangent frame V at Y, and write $\mathrm{D}Y = \mathsf{V}R_0$. Both R_0 and its inverse are uniformly bounded. Normalize columns by R_0^{-1}, so

$$\mathrm{D}X_hR_0^{-1} = \mathsf{V}A + \nu b^{\mathsf{T}}, \qquad N = \frac{\nu - \mathsf{V}A^{-\mathsf{T}}b}{\sqrt{1+|A^{-\mathsf{T}}b|^2}}.$$

Here $A-I$ is the tangential differential perturbation and b the normal slope. After subtracting the first variation, the remainder $\mathcal{R}$ satisfies

$$|P_\nu\mathcal{R}| \le C(|A-I||b| + |b|^3), \qquad |\nu\cdot\mathcal{R}| \le C|b|^2, \quad P_\nu = I - \nu\nu^{\mathsf{T}}. \tag{6.33}$$

Tangential projection removes $m\nabla\psi$ up to an $O(h^3)\nabla\psi$ error. Hence

$$\|A-I\|_2 \le C(D+E), \quad \|b\|_\infty \le C\rho, \quad \|b\|_2 \le C(D+h^{-1}F), \quad \|\nu\cdot z\|_2 \le Ch^2(D+E)+Ch^3E.$$

The tangential remainder pairing is bounded by $C\rho E(D+E) + C\rho^2E(D+h^{-1}F)$; the normal pairing is at most $C\rho(D+h^{-1}F)[h^2(D+E)+h^3E]$. Use $F \le CE$ and $\rho^2/h \le Ch^{1/2} \to 0$. Young's inequality gives (6.32).

The multiplier term retains a particular normal-error factor. From (6.14), (6.18), and (6.12),

$$\begin{aligned}|(\mu,(\mathcal{U}-\mathcal{T})z)_h| \le C\sqrt{\ell}(h^{-2}F+\alpha h^{-1}D+E+h^2)\\ \cdot(h^{-1}F+E+\alpha D+h^3)(D+E).\end{aligned} \tag{6.34}$$

Its leading product is bounded by

$$C\sqrt{\ell}h^{-3}F^2(D+E)\le \vartheta D^2+CE^2+C_\vartheta \ell h^{-6}F^4. \tag{6.35}$$

To account for every other term, the product of the first two parentheses in (6.34), less $h^{-3}F^2$, is exactly

$$\begin{aligned}(h^{-2}+h^{-1})FE+2\alpha h^{-2}FD+(\alpha h^{-1}+\alpha)ED+\alpha^2h^{-1}D^2+E^2\\ +2hF+2\alpha h^2D+(h^2+h^3)E+h^5.\end{aligned}$$

After multiplication by $D+E$, the first eight groups have respective bounds by D^2+E^2 with coefficients

$$Ch^{3/4},\quad Ch^{3/2},\quad Ch^{5/2},\quad Ch^{9/4},\quad Ch^{11/4},\quad Ch,\quad Ch^{11/4},\quad Ch^2.$$

For the fourth group the bound uses only D^2. These follow from $F\le CE$, $D\le Ch^{-1}E$, and (6.13). Each coefficient still tends to zero after multiplication by $\sqrt{\ell}$. The final source gives $\vartheta D^2+CE^2+C_\vartheta h^{10}\ell$. Absorbing the small terms in (6.31) proves

$$(Z^2)'+cD^2\le CE^2+Ch^6+C\ell h^{-6}F^4,\qquad c>0. \tag{6.36}$$

The fourth power here is F^4, rather than E^4.

6.5 Closing the estimates and continuing to the final time

Add a second bootstrap $\mathcal{B}\le Mh^6$, with M fixed independently of h. At time zero $e=0$, since all initial nodes are on Γ^0, and $\mathcal{B}(0)\le Ch^8$. Thus both bootstraps hold strictly on a nonempty initial interval. By (6.29),

$$F^2\le CMh^6+Ch^{15/2}+Ch^9+Ch^8\le C(M+1)h^6.$$

Insert this into (6.36) and use (6.9). The integrating factor e^{-Ct} yields

$$\sup_{s\le t}E(s)^2+\int_0^t D(s)^2\,ds\le C_{M,T}h^6\ell. \tag{6.37}$$

Then (6.30) and $\alpha^2\le Ch^{3/2}$ give

$$\mathcal{B}(t)\le C_Th^6+C_{M,T}h^{3/2}h^6\ell. \tag{6.38}$$

The first C_T is independent of M: it comes from the linear bulk equation, the initial bulk, and the smooth Ch^6 source. Choose $M\ge 4C_T$ first, and then choose h small enough that the second term is at most C_Th^6. This gives $\mathcal{B}\le Mh^6/2$. Also (6.37) gives $E\le h^{11/4}/2$, since $h^{1/4}\sqrt{\ell}\to 0$. Both bootstraps improve strictly.

For completeness, full-time existence is part of this argument. Where element cross products are nonzero and $\mathcal{S}_N$ is invertible, the ORIGINAL system is the smooth finite-dimensional ODE

$$\begin{aligned}X_{h,t}&=-Q_NL_hX_h+\mathcal{U}^*\mathcal{S}_N^{-1}\Pi_h[N\cdot u(X_h,t)],\\ \kappa_h&=\mathcal{S}_N^{-1}\{\Pi_h[N\cdot u(X_h,t)]+\mathcal{U}L_hX_h\}.\end{aligned} \tag{6.39}$$

The initial Schur bound gives local existence and uniqueness: on a small fixed-h nodal ball the vector field has finite value and derivative bounds, and successive integral iterates have factorially summable differences. The same difference estimate proves uniqueness. No Lipschitz constant uniform in h is required.

On its maximal safe interval, the improved estimates imply

$$\left\|X_h - \overline{X}\right\|_{W^{1,\infty}} \le C(h^3 + h^{-2}E) \le C_T h\sqrt{\ell} \longrightarrow 0.$$

The first bootstrap alone already bounds this by $Ch^{3/4}$. The exact embedding has a uniform differential lower bound and local graph radius on compact space-time. On each smaller graph chart, project to its tangent plane and integrate the derivative difference along line segments. Small piecewise $W^{1,\infty}$ perturbations preserve a local lower Lipschitz bound. Continuous FE traces make these restrictions absolutely continuous across edges. Parameter pairs outside these charts have a positive exact-image separation by compactness; uniform C^0 closeness preserves it. Consequently the numerical map remains an embedding with nondegenerate elements. This also justifies the normal graph used on the safe interval.

For fixed h, the bounded nodal positions, positive element Jacobian lower bounds, and (6.15) keep the trajectory in a compact subset of the ODE domain. The vector field is bounded there, so a finite endpoint has a limiting nodal value from which local existence restarts. Strict bootstrap improvement excludes an endpoint before T. Equation (6.39) also gives uniqueness of the velocity and multiplier.

Finally the original closest lift and (6.3), (6.37) give

$$\left\|X_h^\ell - X\right\|_{L^2(\Gamma^0)} \le CE + Ch^4 \le C_T h^3\sqrt{\ell}.$$

The signed-distance estimate follows from (6.21), (6.38). All endpoint estimates hold by continuity. This proves Theorem 6.1.

Remark 6.2. These results partially solve the low-degree question: degree one fails for an admissible family, and degree three converges for every original admissible family under the smooth-solution hypothesis. Degree two remains unresolved. Even substituting a putative $F = O(h^2)$ in (6.36) gives only the driving bound $O(h^2\ell)$, hence a map bound $O(h\sqrt{\ell})$. The inverse estimate then cannot exclude local element degeneration. This failure of the present estimate is not a quadratic counterexample.

References

[1] G. Gao, B. Li and R. Tang, *Convergent finite element approximations of surface evolution with relaxed minimal deformation*, Numerische Mathematik **158** (2026), 671–714. doi:10.1007/s00211-026-01529-3. Author manuscript: https://libuyang.com/NM-15.pdf.

[2] T. Tao, *245C, Notes 2: The Fourier transform*, lecture notes, 6 April 2009, Theorem 43. https://terrytao.wordpress.com/2009/04/06/the-fourier-transform/.

AI-use statement

This work was developed with substantial assistance from OpenAI Codex in mathematical exploration, algebraic checking and manuscript preparation.